

Beyond the conjectured bound

Sharp discrepancy for two families of binary substitutions

Counterexamples and corrected theorems for
OEIS A284365 and A284366

with Part II: exact position-error envelopes
for quadratic morphic words (OEIS A284368–A284371)

and Part III: exact fluctuation laws and sharp
critical and supercritical normalization

Research study prepared with ChatGPT

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Abstract

The OEIS entries A284365 and A284366 conjecture that two complementary position sequences associated with the substitution $0 \mapsto 1, 1 \mapsto 101010$ have linear-approximation errors smaller than 2. Both upper bounds are false. Their first counterexamples occur at ranks 8 933 313 and 2 977 771, respectively, and have the same error 2.0048721733261288... We give exact integer certificates for these values and for their minimality. More substantially, we determine the complete discrepancy limit sets for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, $m \geq 1$. For the OEIS case $m = 3$, the sharp common upper bound is $(6 + \sqrt{21})/5 = 2.116515138991168...$. The proof uses two state-dependent prefix-weight intervals and a contracting set recursion. We also prove the companion density conjectures, construct explicit infinite families of counterexamples, derive rational generating functions for their indices, and provide exact-arithmetic random-access and verification software. The mathematical proofs are self-contained; finite experiments are supporting checks, not substitutes for the proofs.

Part II, added on October 1, 2026 (batch 73O1), applies the same method to the disjoint family $0 \mapsto 1, 1 \mapsto 10^a 1^b$ with $a, b \geq 1$. It proves that prefix discrepancy is bounded exactly when $a \leq b + 1$ (logarithmic growth at $a = b + 2$, power growth beyond), determines the four exact state extrema and the sharp, unattained position-error envelopes throughout the bounded regime, and specializes them to the OEIS words A284368 ($1 \mapsto 1011$) and A284369 ($1 \mapsto 1001$). For A284369 this proves the conjectured bounds of A284370 and A284371 with sharper constants; for A284368 it recovers, as a calibration, the known s -Wythoff floor formulas of A184484–A184485.

Part III, added on October 6, 2026 (batch 114), answers Part II's question on the unbounded regimes. At $a = b + 2$ the discrepancy $\#0 - \#1$ divided by $\log N$ has the cluster interval $[-(a - 1)/(2 \log a), (a - 1)/(2 \log a)]$, with explicit first extremal prefixes; for $a \geq b + 3$ the eigenweight $\#0 - q\#1$ divided by $N^{\log_a q}$ has a closed-form symmetric cluster interval and a continuous graph-directed phase profile (endpoints $\pm\sqrt{10}$ and the bound $X(N)^2 + 5X(N) \leq 10N$ at every prefix for $1 \mapsto 10^b 1$). For every word with a zeros and $a - 1$ ones beginning with 1, an exact positive convolution formula for complete-block prefix histograms yields a random-sum law, exact moments, Gaussian, local and large-deviation laws, sharp variance and pressure comparisons, and a classification of the zero-height profiles, counted by $(a - 1)2^{a-2}$.

Status and scope. The cited OEIS pages still labeled the relevant statements as conjectures when accessed on September 19, 2026. This article establishes the stated results by explicit proofs and executed certificates. It does not claim an exhaustive priority search, peer review, or proof-assistant formalization. General spectral methods for substitution discrepancy are classical; the contribution here is the exact solution and certificates for this family and these particular conjectures.

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Part I. Sharp discrepancy for a family of binary substitutions

1 The selected problem and its resolution

Let σ be the substitution

$$\sigma(0) = 1, \quad \sigma(1) = 101010. \quad (1.1)$$

Its fixed word beginning with 1 is

$$\mathbf{w} = 101010110101011010101 \dots$$

This is A284364 [1]. Number positions starting at 1. Let $u(n)$ and $v(n)$ be the positions of the n th zero and the n th one, respectively. Thus u is A284365 and v is A284366 [2, 3]. Put

$$r = \frac{9 + \sqrt{21}}{6}, \quad s = \frac{\sqrt{21} - 1}{2}, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (1.2)$$

The position-sequence entries conjecture

$$-2 < rn - u(n) < 2, \quad -1 < sn - v(n) < 2 \quad (n \geq 1). \quad (1.3)$$

They also conjecture their respective densities to be $(9 - \sqrt{21})/10$ and $(1 + \sqrt{21})/10$. The density comments are dated April 13, 2025; the sequence entries credit Clark Kimberling, March 26, 2017. We use the unambiguous position-sequence formulations: the parent entry's comment uses count terminology alongside position slopes, which would be a different and immediately incompatible statement.

Theorem 1.1 (Resolution for the OEIS sequences). Write $E_0(n) = rn - u(n)$ and $E_1(n) = sn - v(n)$. For every $n \geq 1$,

$$-\frac{12 + 2\sqrt{21}}{15} < E_0(n) < \frac{6 + \sqrt{21}}{5}, \quad (1.4)$$

$$\frac{\sqrt{21} - 9}{5} < E_1(n) < \frac{6 + \sqrt{21}}{5}. \quad (1.5)$$

Each endpoint is a subsequential limit, and indeed the full set of subsequential limits of each error sequence is the corresponding closed interval. In particular, both conjectured lower bounds in (1.3) hold, but both upper bounds fail infinitely often.

The first counterexamples are

$$v(2\,977\,771) = 5\,334\,043, \quad u(8\,933\,313) = 20\,222\,898. \quad (1.6)$$

Their common error is

$$\frac{2\,977\,771\sqrt{21} - 13\,645\,857}{2} = 2.00487217332612884794005732278\dots > 2. \quad (1.7)$$

The two asserted densities are correct.

The bounds are not consequences of extrapolating the observed maxima. Sections 2–5 prove them for an entire family. Section 6 supplies a finite certificate for (1.6) and its minimality. Section 7 gives an explicit sequence of errors increasing to the sharp upper endpoint. The method belongs to the established study of the relation between substitution spectra and discrepancy [4]; all specific interval and sharpness arguments needed here are proved below.

2 A parameterized family and its discrepancy theorem

Fix an integer $m \geq 1$ and define

$$\sigma_m(0) = 1, \quad \sigma_m(1) = (10)^m. \quad (2.1)$$

The image of 1 begins with 1 and has length at least 2, so the words $\sigma_m^k(1)$ form nested prefixes of an infinite fixed word $\mathbf{w}^{(m)}$. The substitution contains both letters in an iterate of either letter; no additional initial condition is needed beyond the initial 1.

Throughout the general argument let

$$q = \frac{\sqrt{m^2 + 4m} - m}{2}, \quad \lambda = m + q = \frac{m}{q}, \quad s = 1 + q, \quad r = 1 + \frac{1}{q}, \quad (2.2)$$

and

$$t = 1 - q, \quad C = \frac{q}{1 - q^2}, \quad H = qC = \frac{q^2}{1 - q^2}. \quad (2.3)$$

The quantities depend on m , even when the subscript is suppressed. The following identities will be used repeatedly:

$$0 < q < 1, \quad q^2 = m(1 - q), \quad q(1 + H) = C, \quad H(1 + q) = m, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (2.4)$$

For example, $H(1 + q) = q^2/(1 - q) = m$. Also q is irrational: $(m + 1)^2 < m^2 + 4m < (m + 2)^2$ for $m \geq 1$.

Let $Z(N)$ and $O(N)$ count the zeros and ones in the first N letters, with $Z(0) = O(0) = 0$. Let $u_m(n)$ and $v_m(n)$ denote the corresponding position sequences, and set

$$E_0(n) = rn - u_m(n), \quad E_1(n) = sn - v_m(n).$$

Theorem 2.1 (Sharp error intervals for every m). *For every integer $m \geq 1$ and $n \geq 1$,*

$$C - m < E_1(n) < C, \quad (2.5)$$

$$\frac{1 - C}{q} < E_0(n) < C. \quad (2.6)$$

The closures of the two error-value sets are exactly these intervals with their endpoints included. These are also their full subsequential limit sets. No endpoint is attained at a finite index.

For every $N \geq 0$,

$$-\frac{H}{1 + q} < O(N) - \frac{N}{1 + q} < \frac{C}{1 + q}. \quad (2.7)$$

Again, the two endpoints are sharp, and the closure of the centered counting errors is the full closed interval. In particular,

$$\lim_{N \rightarrow \infty} \frac{O(N)}{N} = \frac{1}{s}, \quad \lim_{N \rightarrow \infty} \frac{Z(N)}{N} = \frac{1}{r}.$$

A useful equivalent form of the lower zero-position endpoint is

$$\frac{1 - C}{q} = C - 1 - \frac{m - 1}{q}. \quad (2.8)$$

For $m = 3$, substituting $q = (\sqrt{21} - 3)/2$ gives

$$C = \frac{6 + \sqrt{21}}{5}, \quad H = \frac{3 + 3\sqrt{21}}{10},$$

and (2.5)–(2.6) become (1.5)–(1.4).

3 A contracting weight and two prefix states

For a finite binary word P , define its weight by

$$X(P) = |P|_0 - q|P|_1. \quad (3.1)$$

This is additive under concatenation. The incidence matrix of the substitution, with row order 0, 1 and column order 0, 1, is

$$M = \begin{pmatrix} 0 & m \\ 1 & m \end{pmatrix}.$$

Its eigenvalues are λ and $-q$. More directly, the letter weights give the identity on which the proof rests:

Lemma 3.1 (Weight contraction). *For every finite word P ,*

$$X(\sigma_m(P)) = -qX(P). \quad (3.2)$$

Proof. For $P = 0$ the two sides equal $-q$. For $P = 1$, the image has m letters of each type and hence weight $m(1 - q) = q^2 = -q(-q)$. Additivity proves the claim for an arbitrary concatenation of letters. $\square$

The next letter is crucial. For $a \in \{0, 1\}$, let

$$S_a = \{X(w_1 \cdots w_N) : N \geq 0, w_{N+1} = a\}. \quad (3.3)$$

Thus $0 \in S_1$, from the empty prefix. A prefix ending immediately before a zero belongs to state 0, not to a state determined by its last letter.

Lemma 3.2 (Exact state equations). *The sets in (3.3) satisfy*

$$S_0 = \bigcup_{j=0}^{m-1} (-qS_1 + jt - q), \quad (3.4)$$

$$S_1 = (-qS_0) \cup \bigcup_{j=0}^{m-1} (-qS_1 + jt). \quad (3.5)$$

Proof. View the fixed word as the concatenation of images of its own letters. If a parent prefix P is followed by 0, its image is followed by the single letter 1, giving the transition $-qS_0$ in (3.5). If P is followed by 1, its image is followed by $(10)^m$. Immediately before the $(j + 1)$ st one in that image, the internal prefix is $(10)^j$, whose weight is jt . Immediately before the corresponding zero, the internal prefix is $(10)^j1$, whose weight is $jt - q$. Lemma 3.1 gives all the stated affine images. Conversely, every location in the fixed word lies in one of these image blocks. Thus there are no missing transitions and the equations are set equalities. $\square$

Using a single state and adding all possible offsets independently would forget which parent letter permits which offset. The two-state system retains precisely the information required for the sharp constants.

4 Solving the interval recursion, including sharpness

Define two closed intervals

$$I_0 = [-C, H - 1], \quad I_1 = [-qH, H]. \quad (4.1)$$

They are nondegenerate for every $m \geq 1$.

Lemma 4.1 (Invariant intervals). *The pair (I_0, I_1) satisfies (3.4)–(3.5) with S_a replaced by I_a .*

Proof. Put $J_j = -qI_1 + jt$. Each J_j has length

$$qH(1 + q) = mq,$$

while successive left endpoints are separated by $t = 1 - q = q^2/m$. If $m \geq 2$, then $mq \geq t$, so adjacent intervals overlap. For $m = 1$ there is only one interval. The extreme endpoints yield

$$\bigcup_{j=0}^{m-1} J_j = [-qH, H - t]. \quad (4.2)$$

Indeed, the upper endpoint is $q^2H + (m - 1)t = H - t$, since $q^2(H + 1) = H$ and $mt = q^2$.

Subtracting q from (4.2) gives

$$\bigcup_{j=0}^{m-1} (J_j - q) = [-qH - q, H - t - q] = [-C, H - 1] = I_0.$$

For state 1, the other image is

$$-qI_0 = [q - qH, H].$$

It overlaps or touches the interval in (4.2), because

$$H - t \geq q - qH \iff H(1 + q) \geq 1 \iff m \geq 1.$$

Their union is therefore $[-qH, H] = I_1$. □

Lemma 4.2 (Strict inclusion of all actual prefixes). *Every element of S_a lies in the interior of I_a .*

Proof. We induct on the length N of the prefix. The empty prefix has weight 0, which lies strictly inside I_1 .

For a nonempty prefix, locate its next letter in an image block. The associated parent prefix has some length $k < N$. To see strict decrease, if $k > 0$ then that parent prefix begins with 1, so its image has length at least $k + (2m - 1) > k$; this image is contained in the original prefix. If $k = 0$, strict decrease follows from $N > 0$.

By induction, the parent weight belongs to the interior of its appropriate interval. Its actual transition is an affine map with nonzero slope $-q$. The image of an interior point is an interior point of the image interval, and Lemma 4.1 puts that image interval inside the appropriate target interval. Such a point cannot be an endpoint of the target interval. This completes the induction. □

Bounds alone do not prove sharpness. We next show that the intervals contain no gap in the closures of the actual prefix weights.

Lemma 4.3 (Uniqueness for the compact set recursion). *Let K_0, K_1 be nonempty compact subsets of $\mathbb{R}$ satisfying the state equations. Then $K_0 = I_0$ and $K_1 = I_1$.*

Proof. For compact nonempty subsets of $\mathbb{R}$, write d_H for Hausdorff distance. An affine map of slope $-q$ multiplies this distance by q . Also, for finite indexed families,

$$d_H\left(\bigcup_i A_i, \bigcup_i B_i\right) \leq \max_i d_H(A_i, B_i).$$

Let $D = \max\{d_H(K_0, I_0), d_H(K_1, I_1)\}$. Apply these observations to the two state equations for K and I . They imply $D \leq qD$. Since D is finite and $0 < q < 1$, it follows that $D = 0$. $\square$

Proposition 4.4 (Exact closures).

$$\text{cl } S_0 = [-C, H - 1], \quad \text{cl } S_1 = [-qH, H]. \quad (4.3)$$

The closure of all prefix weights is $[-C, H]$.

Proof. Lemma 4.2 makes S_0, S_1 bounded, and both are nonempty. Take closures in their exact finite-union equations. Closure commutes with each affine map and with a finite union, so $(\text{cl } S_0, \text{cl } S_1)$ is a compact fixed pair. Lemma 4.3 proves (4.3). Finally I_0 and I_1 overlap or touch, since $H - 1 \geq -qH$ is again equivalent to $m \geq 1$. Their union is $[-C, H]$. $\square$

Distinct prefix lengths have distinct weights. Indeed, equality would give $\Delta Z = q\Delta O$, impossible for irrational q unless both integer differences vanish, in which case the lengths agree. A set dense in a nondegenerate interval has infinitely many distinct points in every relative neighborhood of every point of that interval. Consequently its density statements remain valid after deleting any finite number of prefixes. This proves that the closures just obtained are also the full sets of limits along prefix lengths tending to infinity.

5 Position errors, densities, and exact transfer

5.1 Passing from prefix weights to position errors

Immediately before the n th one, the prefix has $n - 1$ ones and $v_m(n) - n$ zeros. Its weight is therefore

$$X = v_m(n) - n - q(n - 1) = q - E_1(n). \quad (5.1)$$

Immediately before the n th zero, the prefix has $n - 1$ zeros and $u_m(n) - n$ ones, giving

$$X = n - 1 - q(u_m(n) - n) = qE_0(n) - 1. \quad (5.2)$$

Apply the strict state bounds and Proposition 4.4. Equation (5.1) transforms I_1 into

$$[q - H, q + qH] = [C - m, C],$$

and (5.2) transforms I_0 into $[(1 - C)/q, C]$. The identities follow from (2.4). The state sets contain exactly the prefixes immediately before the respective ranked letters, so no additional selection restriction is needed. This proves the position part of Theorem 2.1, including strictness and the full subsequential limit sets.

For ordinary prefix counts,

$$X = N - (1 + q)O(N), \quad O(N) - \frac{N}{1 + q} = -\frac{X}{1 + q}. \quad (5.3)$$

The total prefix-weight interval $[-C, H]$ proves (2.7) and its sharpness. Dividing the bounded counting error by N establishes the densities without invoking an ergodic theorem. In the case $m = 3$, this yields the quantitative estimate

$$-\frac{33 + 3\sqrt{21}}{50} < O(N) - \frac{1 + \sqrt{21}}{10}N < \frac{27 + 7\sqrt{21}}{50}. \quad (5.4)$$

Thus the density conjectures in all three relevant OEIS entries follow, and the error in the unnormalized counting function is uniformly bounded.

5.2 A relation between the two position sequences

Proposition 5.1 (Exact position transfer). *For $n \geq 1$ and $1 \leq h \leq m$,*

$$u_m(m(n-1) + h) = v_m(n) + (2m-1)n - 2m + 2h. \quad (5.5)$$

Equivalently,

$$E_0(m(n-1) + h) = E_1(n) - (m-h)(r-2). \quad (5.6)$$

In particular,

$$u_m(mn) = v_m(n) + (2m-1)n, \quad E_0(mn) = E_1(n). \quad (5.7)$$

Proof. Before the parent word's n th one there are $n-1$ ones and $v_m(n) - n$ zeros. The total length of their images is $2m(n-1) + v_m(n) - n$. The image of the next one contains m zeros, at internal positions $2, 4, \dots, 2m$, and no zero comes from the image of a parent zero. Thus the h th such zero has global rank $m(n-1) + h$ and the position in (5.5).

Subtract this expression from $r(m(n-1) + h)$. The identity $m(r-2) = q$ reduces the result to (5.6). $\square$

For $m = 3$, $C < r$; explicitly

$$r - C = \frac{9 - \sqrt{21}}{30} > 0.$$

Since $r - 2 > 0$, (5.6) shows that the errors at zero ranks $3(n-1) + 1$ and $3(n-1) + 2$ are strictly less than $C - (r-2) < 2$. It follows that

$$\{k : E_0(k) > 2\} = \{3n : E_1(n) > 2\}. \quad (5.8)$$

Equality to 2 is impossible, because r and s are irrational. This bijection will make the zero-position minimality result immediate once the first one-position violation has been certified.

6 An exact finite certificate and the first counterexamples

6.1 Recursive blocks

Define auxiliary finite words

$$W_{-1} = 0, \quad W_0 = 1, \quad W_k = (W_{k-1}W_{k-2})^m \quad (k \geq 1). \quad (6.1)$$

For $k \geq 0$, induction gives $W_k = \sigma_m^k(1)$. In particular, W_k is a prefix of $\mathbf{w}^{(m)}$. Write L_k, Z_k, O_k for its length, zero count, and one count. Each satisfies

$$a_k = m(a_{k-1} + a_{k-2}). \quad (6.2)$$

For the remainder of this section $m = 3$.

Consider the finite word

$$P = W_{11} W_{10} W_{10} W_8 W_6 W_4 W_2. \quad (6.3)$$

This is a prefix of W_{12} , followed by a 1. Here is an explicit way to follow the recursive decomposition. In $W_{12} = (W_{11}W_{10})^3$, take the first $W_{11}W_{10}$, then enter the next $W_{11} = (W_{10}W_9)^3$ and take its first W_{10} . The residual word begins with W_9 . In $W_9 = (W_8W_7)^3$, take W_8 , leaving a word beginning with W_7 . Repeat with W_7, W_5, W_3 , taking W_6, W_4, W_2 . The remaining word begins with $W_1 = 101010$, so its next letter is 1.

The count recurrence gives

$$|P|_0 = 2\,356\,272, \quad |P|_1 = 2\,977\,770, \quad |P| = 5\,334\,042. \quad (6.4)$$

Consequently the next one proves the first identity in (1.6). Formula (5.7) proves the second:

$$u(3 \cdot 2\,977\,771) = 5\,334\,043 + 5 \cdot 2\,977\,771 = 20\,222\,898.$$

Their common error is exactly (1.7). To certify that it exceeds 2, it suffices to observe

$$21(2\,977\,771)^2 - (13\,645\,861)^2 = 265\,940 > 0. \quad (6.5)$$

Both quantities compared before squaring are positive, so (6.5) proves $2\,977\,771\sqrt{21} > 13\,645\,861$ with no numerical approximation.

6.2 Why no earlier index is a counterexample

For any finite word U , define

$$\mu_1(U) = \min\{X(Q) : Q \text{ is a proper prefix of } U, \text{ and the next letter in } U \text{ is } 1\}.$$

Use $+\infty$ if U has no one. For concatenations,

$$\mu_1(UV) = \min\{\mu_1(U), X(U) + \mu_1(V)\}, \quad (6.6)$$

$$\mu_1(B^m) = \mu_1(B) + \min\{0, (m-1)X(B)\}. \quad (6.7)$$

The second formula follows by considering the m translated copies of B . These identities give a small exact certificate for all earlier positions at once.

Apply (6.6) to the seven factors of P . The translated candidate minima, in their order of appearance, are

$$\begin{aligned} &431\,712 - 545\,583q, & 1\,820\,700 - 2\,300\,934q, \\ &2\,222\,379 - 2\,808\,561q, & 2\,346\,957 - 2\,965\,998q, \\ &2\,355\,624 - 2\,976\,951q, & 2\,356\,227 - 2\,977\,713q, \\ &2\,356\,269 - 2\,977\,766q. \end{aligned} \quad (6.8)$$

Their minimum is the first, so

$$\mu_1(P) = 431\,712 - 545\,583q. \quad (6.9)$$

All of these finite comparisons can be checked by the integer sign rule in Section 8; the separate checker in the archive recomputes the block table and its minimum rather than accepting them as input.

k	L_k	Z_k	O_k	$\mu_1(W_k)$
0	1	0	1	0
1	6	3	3	0
2	21	9	12	$6 - 8q$
3	81	36	45	$9 - 12q$
4	306	135	171	$99 - 126q$
5	1 161	513	648	$144 - 183q$
6	4 401	1 944	2 457	$1 440 - 1 821q$
7	16 686	7 371	9 315	$2 088 - 2 640q$
8	63 261	27 945	35 316	$20 718 - 26 184q$
9	239 841	105 948	133 893	$30 033 - 37 956q$
10	909 306	401 679	507 627	$297 819 - 376 374q$
11	3 447 441	1 522 881	1 924 560	$431 712 - 545 583q$
12	13 070 241	5 773 680	7 296 561	$4 280 832 - 5 409 957q$

Table 1: Block data for $m = 3$. Every minimum is calculated by (6.6)–(6.7), using exact quadratic-field comparisons. The $k = 12$ row is included for context; only the preceding rows are needed to certify P .

By (5.1), the largest one-position error strictly before the candidate equals

$$q - \mu_1(P) = 545\,584q - 431\,712 = 1.988978393507077166 \dots < 2. \quad (6.10)$$

The last inequality has the exact certificate

$$(2\,500\,180)^2 - 21(545\,584)^2 = 110\,224 > 0. \quad (6.11)$$

It proves $545\,584\sqrt{21} < 2\,500\,180$, which is precisely (6.10) < 2 after substituting the value of q . The maximum in (6.10) occurs at rank 545 584, at position 977 296.

Every earlier ranked one lies in P . Hence no earlier one-position error is at least 2. Equations (6.5) and (6.11) certify the first counterexample, and the exact bijection (5.8) certifies the first zero-position counterexample. This proof covers all earlier indices; it is not a claim based only on a finite sample of selected positions.

7 Infinite extremal families and their generating functions

The compact-set proof gives sharpness nonconstructively in terms of closures. A specific family approaching the common upper endpoint can also be written down and evaluated by short recurrences.

Proposition 7.1 (Explicit extremal prefixes). *For general $m \geq 1$, let P_0 be the empty word and define*

$$P_{j+1} = \sigma_m^2(P_j)(10)^m, \quad U_j = \sigma_m(P_j). \quad (7.1)$$

Every P_j and every U_j is a prefix of the fixed word, followed by a 1. Set

$$n_j = |U_j|_1 + 1, \quad p_j = |U_j| + 1. \quad (7.2)$$

Then

$$v_m(n_j) = p_j, \quad E_1(n_j) = C(1 - q^{2j+2}) \nearrow C. \quad (7.3)$$

The zero-position errors at ranks mn_j are equal to these errors.

Proof. Suppose P_j is a prefix followed by 1. Its image is followed by $(10)^m$. Therefore $\sigma_m(P_j)1$ is a prefix followed by 0. Apply the substitution again: the resulting prefix is $\sigma_m^2(P_j)(10)^m = P_{j+1}$, followed by $\sigma_m(0) = 1$. This proves the prefix assertion inductively from P_0 . It also shows that U_j is followed by 1.

By Lemma 3.1 and $m(1 - q) = q^2$,

$$X(P_{j+1}) = q^2 X(P_j) + q^2.$$

Starting at zero and summing the geometric progression gives $X(P_j) = H(1 - q^{2j})$. Thus

$$E_1(n_j) = q - X(U_j) = q + qH(1 - q^{2j}) = C - Cq^{2j+2},$$

where $q + qH = C$ and $qH = Cq^2$. The errors increase strictly because $0 < q < 1$. The zero-position assertion follows from (5.7). $\square$

Let $(z_j, o_j) = (|P_j|_0, |P_j|_1)$. The preceding construction is fully integer-recursive:

$$\begin{pmatrix} z_{j+1} \\ o_{j+1} \end{pmatrix} = \begin{pmatrix} m & m^2 \\ m & m^2 + m \end{pmatrix} \begin{pmatrix} z_j \\ o_j \end{pmatrix} + \begin{pmatrix} m \\ m \end{pmatrix}, \quad (z_0, o_0) = (0, 0), \quad (7.4)$$

with

$$n_j = z_j + mo_j + 1, \quad p_j = z_j + 2mo_j + 1. \quad (7.5)$$

Eliminating the two counts yields an especially simple result.

Corollary 7.2 (Scalar recurrences and rational generating functions). *Put $T = m(m + 2)$. Then*

$$n_{j+2} = Tn_{j+1} - m^2n_j + 1 - m, \quad n_0 = 1, \quad n_1 = m^2 + m + 1, \quad (7.6)$$

$$p_{j+2} = Tp_{j+1} - m^2p_j + m^2 - m + 1, \quad p_0 = 1, \quad p_1 = 2m^2 + m + 1. \quad (7.7)$$

Their ordinary generating functions are

$$\sum_{j \geq 0} n_j z^j = \frac{1 - mz}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}, \quad (7.8)$$

$$\sum_{j \geq 0} p_j z^j = \frac{1 + m(m - 1)z}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}. \quad (7.9)$$

Proof. The linear part in (7.4) is M^2 , with trace T and determinant m^2 . Its characteristic identity implies the two inhomogeneous recurrences after substitution in (7.5); alternatively, one may eliminate z_j, o_j directly. The constant terms and initial values follow by evaluating $j = 0, 1, 2$.

For a sequence obeying $a_{j+2} = Ta_{j+1} - m^2a_j + c$, multiplying its series by $1 - Tz + m^2z^2$ gives $a_0 + (a_1 - Ta_0)z + cz^2/(1 - z)$. For (7.6), the numerator after multiplication by $1 - z$ reduces to $1 - mz$; for (7.7), it reduces to $1 + m(m - 1)z$. This proves both generating functions. $\square$

In the OEIS case the recurrences are $n_{j+2} = 15n_{j+1} - 9n_j - 2$ and $p_{j+2} = 15p_{j+1} - 9p_j + 7$. Formula (7.3) shows that every $j \geq 6$ gives a counterexample. The closed-interval theorem is stronger still: every open interval contained in $(2, C)$ contains infinitely many errors of each position sequence, not just those in this special family.

j	n_j	$p_j = v(n_j)$	$E_1(n_j)$, rounded
0	1	1	0.791287847478
1	13	22	1.286742017213
2	184	328	1.596963935937
3	2 641	4 729	1.791205189187
4	37 957	67 990	1.912826719410
5	545 584	977 296	1.988978393507
6	7 842 145	14 047 537	2.036659732964
7	112 721 917	201 917 398	2.066514757944
8	1 620 249 448	2 902 333 144	2.085208077525

Table 2: An explicit extremal family for $m = 3$. It supplies infinitely many counterexamples, but its first violation ($j = 6$) is not the first violation over all ranks.

7.1 Exact asymptotics along the extremal family

The denominator in (7.8)–(7.9) factors as

$$(1 - z)(1 - \lambda^2 z)(1 - q^2 z).$$

Here $\lambda > 1$ and $0 < q < 1$, so partial fractions give an exact three-term expansion. For either sequence $a_j = n_j$ or $a_j = p_j$, let

$$d = \frac{c}{1 - 2m}, \quad A = \frac{a_1 - d - q^2(a_0 - d)}{\lambda^2 - q^2}, \quad B = a_0 - d - A,$$

where c is the corresponding constant in its recurrence. Then

$$a_j = A\lambda^{2j} + d + Bq^{2j}. \quad (7.10)$$

The leading coefficient is positive, as follows either from these expressions or the positive count recurrence. In particular, $n_j, p_j = \Theta(\lambda^{2j})$, whereas

$$C - E_1(n_j) = Cq^{2j+2} = \Theta(p_j^{-\kappa}), \quad \kappa = -\frac{\log q}{\log \lambda}. \quad (7.11)$$

For $m = 3$, $\kappa = 0.17565278149 \dots$. This is a statement along the constructed subsequence, not a claim about the optimal record-error convergence rate over every prefix. The comparatively small exponent explains why substantial growth in the rank can coexist with a slow approach to the limiting extremum.

8 Exact algorithms and executed checks

8.1 Quadratic arithmetic without rounding decisions

Every prefix weight and every one-position error is of the form $a + bq$ with integer a, b . Since $q^2 = m - mq$, addition and multiplication stay in this representation. To compare $a + bq$ with zero, put

$$A = 2a - mb, \quad D = m^2 + 4m.$$

Its sign is the sign of $A + b\sqrt{D}$. If A and b have the same nonzero sign, that sign is the answer. Zero cases are immediate. If they have opposite signs, the answer is

$$\text{sgn}(A) \text{sgn}(A^2 - Db^2). \quad (8.1)$$

As D is nonsquare, there is no unhandled nontrivial equality case. Thus all block-extremum decisions and counterexample inequalities in the software reduce to integer arithmetic. Zero-position errors may contain $1/q$; multiplying their comparisons by the positive number q puts them in the same representation.

8.2 Associative summaries and random access

A block summary stores its zero and one counts, and for each following letter state the minimum and maximum prefix weight, together with a location attaining the extremum. Under concatenation UV , the extrema from V are translated by $X(U)$ and their positions by $|U|$; the appropriate minimum or maximum is then taken with the extrema from U . This is the same identity as (6.6) in both states and both directions.

Precompute summaries for $W_{-1}, W_0, \dots, W_k$ using (6.1). To count the first N letters, find the first block long enough, skip whole child pairs by their lengths, and descend into the residual child. To find the n th specified letter, use its block counts instead of lengths. At a fixed m , there are $O(\log N)$ levels, and each level involves only a bounded number of arithmetic operations. The library groups repeated child pairs for these operations.

To find the first error at least a threshold B , a block beginning after a prefix of weight x can be discarded exactly when its appropriate extremum cannot reach the threshold. For ones the feasibility condition is

$$x + \mu_1(W_k) \leq q - B.$$

For zeros, using the maximum prefix weight $\nu_0(W_k)$ before a zero, the condition is

$$x + \nu_0(W_k) \geq qB - 1.$$

Find a first feasible enclosing block, then inspect children from left to right and descend into the first feasible child, skipping the others by their counts. The returned leaf is the first violation, and the skipped block extrema certify that no earlier one was missed.

For fixed m and a counterexample at a position of size N , the precomputation and descent use $O(\log N)$ arithmetic operations on $O(\log N)$ -bit integers. This is an arithmetic-operation count, not a unit-cost bit-complexity assertion. With $M(b)$ denoting the cost of multiplying b -bit integers, the sign tests give the conservative bound $O(\log N M(\log N))$ bit operations, alongside additions and small-parameter divisions. Storing all summaries uses $O((\log N)^2)$ bits at fixed m . Searching only up to a prescribed level and finding nothing is a finite negative result, not a proof of nonexistence in the infinite word.

8.3 What was actually checked

The delivered script code `verify.py` -long completed successfully under Python 3.13.5. It recorded 310 816 explicit checks, in addition to the assertions internal to the separate certificate checker. For every $m = 1, \dots, 8$, it built a literal 12 000-letter prefix, checked the sharp bounds at every letter, and compared selected random-access queries and complete small-block extrema against direct calculations. It also checked the transfer identity, the exact extremal family through $j = 24$, and the proposed recurrences and generating-function numerators.

For $m = 3$, the script compared the first 30 displayed terms of each position sequence and the first 30 binary letters with the OEIS entries. It did not download or compare the complete linked 10 000-term tables. The optional long test independently formed the fixed-word prefix of length 20 222 898 by literal substitution, confirming both specified counterexample positions and ranks. The first-counterexample search used exact summaries, not a floating-point scan.

The file code/`independent_check.py` does not import the main library. It reconstructs the count and minimum certificate with a separate integer sign test and concatenation implementation. This checks the finite computations supporting minimality; the prefix-membership argument and transfer theorem supply their mathematical justification. The verification data in `data/` records exact coefficients as well as display decimals, and includes the full first-violation descent traces. No experiment is used to infer the infinite interval theorem.

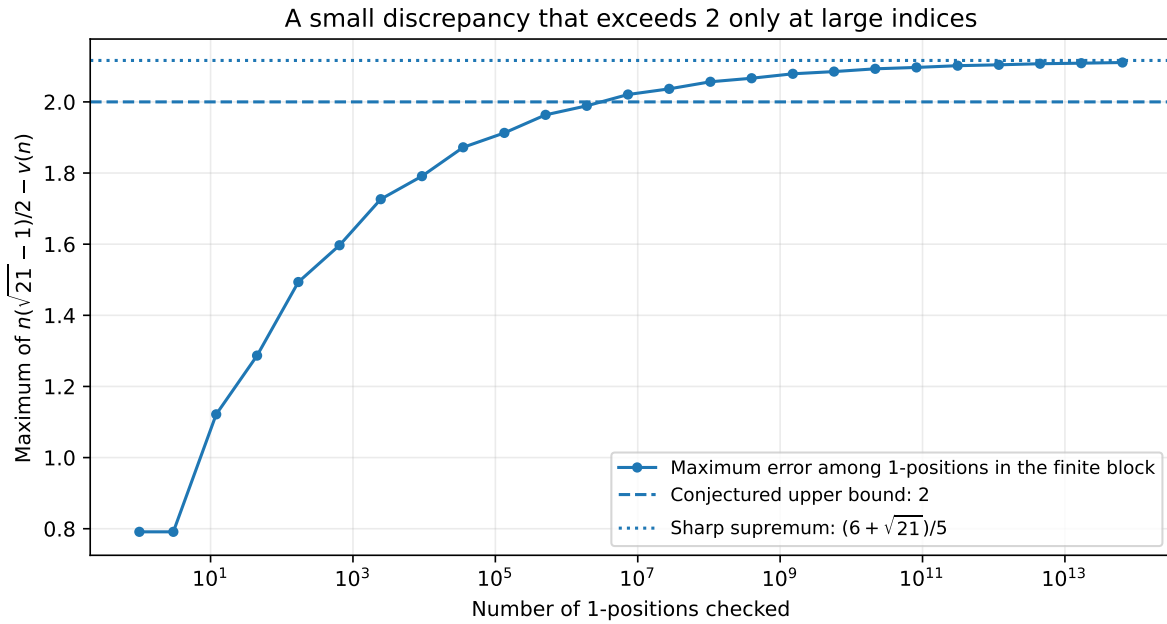


Figure 1: Maximum one-position error over all ones in W_k , for $k = 0, \dots, 24$. The horizontal scale is the number of one-positions in the enclosing block, not the rank at which its maximum is attained. The curve is computed from exact block summaries; floating-point values are used only to render the figure.

9 Further consequences and boundaries of the result

9.1 The $m = 1$ endpoint is a Beatty pair

At $m = 1$, $q = (\sqrt{5} - 1)/2$ and $C = 1$. Both error intervals become $(0, 1)$. Since the positions are integers and the slopes are irrational,

$$v_1(n) = \lfloor (1 + q)n \rfloor, \quad u_1(n) = \left\lfloor \left(1 + \frac{1}{q}\right)n \right\rfloor.$$

Thus the familiar complementary Beatty description is recovered directly from the theorem. For $m \geq 2$, no fixed real shift γ can give $v_m(n) = \lfloor sn + \gamma \rfloor$ for every n : such an expression would put all its errors in an interval of length 1, whereas the error closure in (2.5) has length exactly m . The density already forces any putative Beatty slope to be s .

9.2 A finite factor-balance bound

The centered prefix-counting errors have range width

$$\frac{C + H}{1 + q} = C.$$

For a factor of length L , its number of ones minus L/s is the difference of two prefix errors, so it lies strictly between $-C$ and C . Two factors of the same length consequently differ in their numbers of ones by an integer whose absolute value is strictly less than $2C$. Hence

$$||A|_1 - |B|_1| \leq \lceil 2C \rceil - 1 \quad \text{whenever } |A| = |B|. \quad (9.1)$$

For $m = 3$ this gives a balance bound of 4. The article does not assert that this bound is optimal: sharp prefix extrema do not automatically occur at two endpoints separated by the same prescribed factor length. The exact optimal factor-balance constant is a separate question.

9.3 The word itself has a nonrational generating function

Although the extremal-index series (7.8)–(7.9) are rational, the characteristic series

$$F_m(z) = \sum_{n \geq 1} w_n^{(m)} z^n$$

is not rational. Indeed, a rational power series has coefficients obeying an eventual fixed linear recurrence. If those coefficients take values in a finite set, the vector of a fixed number of consecutive coefficients has only finitely many states and deterministically produces its successor. A state eventually repeats, making the sequence eventually periodic. An eventually periodic binary sequence has rational density, contradicting $1/(1+q)$, which is irrational. This elementary observation concerns the full word, not the rational generating functions of the sparse families.

9.4 What remains outside the claims

The conjectured upper bounds and density statements selected in Section 1 are resolved by the preceding proofs. The article does not determine the frequency distribution of discrepancy values within their interval, the density of the ranks with error above 2, or the exact best factor-balance constant. Knowing the support of subsequential limits does not by itself answer those distributional questions. Nor is the constructed extremal subsequence claimed to enumerate all record errors. These distinctions prevent stronger statements from being inferred merely from the observed finite data or the interval closure result. [Added 1 October 2026, batch 73O1: Part II (research question 3 of Section 21) asks the same density question for the family $0 \mapsto 1$, $1 \mapsto 10^a 1^b$; it remains open in both Parts. The scope of Part I is the family $1 \mapsto (10)^m$ only; Part II treats the disjoint family $1 \mapsto 10^a 1^b$ by the same method (Remark 11.1).]

10 Conclusion

A conjecture about an error below 2 can survive a substantial finite check while failing only slightly at a much larger index. Here the first one-position violation occurs at rank 2 977 771, and its transfer produces the first zero-position violation at rank 8 933 313. The finite disproof is certified by a short recursive prefix and two integer square comparisons. The structural analysis goes further: the correct common upper bound is $(6 + \sqrt{21})/5$, and a two-state contraction proves exact discrepancy intervals for every $0 \mapsto 1$, $1 \mapsto (10)^m$. Density, explicit extremal families, rational index-generating functions, and efficient exact verification then follow from the same framework.

Part II. Exact position-error envelopes for quadratic morphic words

A two-parameter discrepancy theorem with sharp applications to OEIS A284368–A284371

Research study prepared with ChatGPT, dated October 1, 2026; added to this report on 1 October 2026 in batch 7301 of ProveIt's incoming-report intake.

Sections 1–10 and Appendix A form the original report (Part I) of September 19, 2026, unchanged except for dated notes pointing to this addition, a front matter (title and abstract) that now announces both Parts, and the place of Appendix A, which now follows Part II together with Part II's own appendices. Part II was written as an extension of Part I's method to the family $0 \mapsto 1, 1 \mapsto 10^a 1^b$; it neither contains nor re-proves Part I's results. Section 11 records the provenance, fixes the notation of the addition, and compares the two families. [Added 6 October 2026, batch 114: Part III (Sections 23–36) follows Part II and answers its research question 7; the front matter now also announces Part III.]

Summary (the manuscript's abstract). We study the fixed point of the binary substitution

$$\tau_{a,b}(0) = 1, \quad \tau_{a,b}(1) = 10^a 1^b, \quad a, b \geq 1.$$

Let

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2}.$$

The non-Perron eigenvalue of the incidence matrix is $-q$. We prove a sharp phase transition: prefix discrepancy, and hence the errors in the positions of zeros and ones, is bounded exactly when $a \leq b+1$. Throughout that bounded regime we determine the four exact state-dependent extrema. Writing

$$\widehat{C} = \frac{a-q}{1-q^2},$$

the closures of the position-error sets have endpoints

$$q - \widehat{C}, \quad q(1 + \widehat{C})$$

for positions of ones, and

$$\frac{1}{q} - \widehat{C} - 1, \quad \frac{C}{q}$$

for positions of zeros. None of these endpoints is attained, and all are subsequential limits.

The specializations $(a, b) = (1, 2)$ and $(2, 1)$ are OEIS A284368 and A284369. For A284368 we recover the sharper bounds

$$\frac{\sqrt{13}-5}{3} < E_1(n) < \frac{\sqrt{13}-2}{3}, \quad \frac{1+\sqrt{13}}{3} < E_0(n) < \frac{4+\sqrt{13}}{3}.$$

For A284369 we prove

$$-2 < E_1(n) < 1 + \sqrt{3}, \quad -\frac{3+\sqrt{3}}{2} < E_0(n) < 2 + \sqrt{3}.$$

The latter resolves the still-posted conjectures for its position sequences A284370 and A284371, with strictly stronger constants. In both OEIS cases the complete state attractors are intervals,

so the displayed endpoints are not merely estimates but the exact error-value envelopes. We also give constructive extremal subsequences, rational generating functions for their indices, a standard-library exact verifier, and a research program for the remaining interval-versus-Cantor and distribution questions.

Status and claim boundary. This Part extends the method of the existing ProveIt report on $0 \mapsto 1, 1 \mapsto (10)^m$ (Part I) rather than repackaging its A284364–A284366 results. The OEIS pages cited below still presented the A284368 and A284369 position statements as conjectures when checked on October 1, 2026. A284368 is already identified with the generalized s -Wythoff pair A184484–A184485, so its sharp floor formulas have an independent route and are included here as a calibration, not as the main novelty. The principal claims are the two-parameter theorem and the sharp A284369–A284371 envelopes. A limited source search found no proof of the A284369 bounds. No claim of exhaustive literature priority, peer review, or proof-assistant verification is made. The proofs here are self-contained, and the included computations use exact quadratic arithmetic.

11 Provenance, conventions and relation to Part I

Part II is built from one manuscript, number 44 of batch 73O1: *Exact position-error envelopes for quadratic morphic words* (October 1, 2026), delivered as OEIS_Quadratic_Morphic_Words_Research_Package.zip in arrival commit c79d64038 and placed in commit 9df4ba51a. It names no repository commit; it cites Part I by its repository URL. Its programs, data and figures are shipped with the prefix 02-morphic- (Appendix C and the README). The merge made these choices.

- Every theorem, corollary, proposition, lemma, remark, proof, table, figure and research question of the manuscript is printed. Its section n is Section $n + 11$ here, and its Appendices A, B are Appendices B, C.
- The manuscript cited Part I as “V. Reshetnikov, ProveIt repository, . . . , research draft”; Part I’s author line is “Research study prepared with ChatGPT”. The citation is replaced by references to Part I’s sections. The manuscript’s reference to Adamczewski’s 2004 paper is merged with Part I’s.
- The manuscript’s constant C is printed as $\widehat{C}$ (Table II.1); nothing else is renamed and no normalization changed.
- File names in the text are the shipped names; the delivered programs themselves still use their delivery names (Appendix C).

Notation. Part II uses Part I’s definitions of the prefix weight X , the next-letter states S_a , the counting function $O(N)$ and the position errors $E_0(n) = rn - u(n)$, $E_1(n) = sn - v(n)$, for its own word. Where a letter is shared with a different meaning, Table II.1 prints the Part I meaning beside the Part II one.

Remark 11.1 (The two families, compared). *The families are disjoint.* The word $(10)^m$ has the form 10^a1^b only for $m = 1$, with $(a, b) = (1, 0)$, which Part II excludes $(a, b \geq 1)$. So Part II neither contains, re-proves nor refines Part I’s theorems for A284364–A284366, and it does not say it does.

The same quadratic fields occur. For $(a, b) = (m, m - 1)$ the incidence matrix of $\tau_{a,b}$ is $\begin{pmatrix} 0 & m \\ 1 & m \end{pmatrix}$, the matrix of σ_m , so q, λ, r and s coincide; the words and the envelopes differ. For $m = 3$,

Table II.1: Symbols shared by the two Parts.

Symbol	Part I ($\sigma_m: 1 \mapsto (10)^m$)	Part II ($\tau_{a,b}: 1 \mapsto 10^a 1^b$)
q, λ, r, s	(2.2): $q = \frac{\sqrt{m^2+4m-m}}{2}, \lambda = m/q$	(13.3), (13.6): $q = \frac{\sqrt{(b+1)^2+4a-(b+1)}}{2}, \lambda = a/q$; r, s by the same formulas
$\widehat{C}$ vs C	$C = q/(1 - q^2)$ (2.3), the upper end of both position-error intervals	$\widehat{C} = (a - q)/(1 - q^2)$ (13.8), the manuscript's C ; it is the largest state-1 weight, the role played in Part I by H , not by C
H, t	(2.3); $t = 1 - q$	not used (t is a threshold in research question 3 of Section 21)
M	incidence matrix of σ_m	incidence matrix of $\tau_{a,b}$ (13.2)
I_0, I_1	the intervals (4.1)	the candidate hulls in each OEIS case, (17.2), (17.6)
K_0, K_1	any compact solution in Lemma 4.3	$K_i = \text{cl } S_i$ (Section 14)
P_j, P_k	extremal prefixes (7.1)	the different cycle (16.1)
T	$T = m(m + 2)$ (Corollary 7.2)	T_k , a prefix word (Section 18)
u	$u_m(n)$, position of the n th zero	$u(n)$ likewise; $u_i = \max K_i$ in Section 15 is a state extremum, not a position
F	$F_m(z)$, the characteristic series of the word	$F_0(n), F_1(n)$, the A284369 position errors

Part I gives one-position errors in $(C - 3, C) \approx (-0.8835, 2.1165)$ and zero-position errors in $((1 - C)/q, C) \approx (-1.4110, 2.1165)$, while Theorem 13.1 for $(a, b) = (3, 2)$ gives $(q - \widehat{C}, q(1 + \widehat{C})) \approx (-5.1165, 5.4661)$ and $(1/q - \widehat{C} - 1, \widehat{C}/q) \approx (-5.6440, 7.4661)$.

A formal consistency check. Setting $(a, b) = (1, 0)$ in the formulas of Theorem 13.1, outside its hypotheses, gives $q = (\sqrt{5} - 1)/2$, $\widehat{C} = q$, and the interval $(0, 1)$ for both letters: the complementary Beatty pair of Part I's case $m = 1$ (Section 9). Part II's proof does not cover this case (it uses $\widehat{C} - q^2(\widehat{C} + 1) = qb > 0$).

Method. Both Parts rest on the contracting weight $X = |P|_0 - q|P|_1$ with $X(\text{image}) = -qX$ and on states recording the next letter (Part I, Sections 3–4; Part II, Sections 14–15). Part II solves a four-variable Bellman system for the hulls and treats the question whether the attractors fill them separately (Section 17), whereas in Part I the intervals are invariant for every m .

Open questions. Part II's research question 3 (densities of ranks with error above a threshold) is the question Part I leaves open in Section 9; it is not answered in either Part.

12 The OEIS questions and the new result

12.1 Two neighboring morphic words

The fixed point of

$$0 \mapsto 1, \quad 1 \mapsto 1011 \quad (12.1)$$

is A284368. Its entry identifies the positions of ones and zeros with A184484 and A184485, respectively, and conjectures [6, 7, 8]

$$1 < n \frac{5 + \sqrt{13}}{2} - u(n) < 3, \quad -1 < n \frac{\sqrt{13} - 1}{2} - v(n) < 1. \quad (12.2)$$

Here and below $u(n)$ is the position of the n th zero and $v(n)$ the position of the n th one. The older entries A184484–A184485 describe the same complementary pair as an s -Wythoff

pair. Thus some consequences can also be approached through generalized Beatty theory; our purpose is an exact substitution-state analysis that extends uniformly to a larger family.

The neighboring fixed point

$$0 \mapsto 1, \quad 1 \mapsto 1001 \quad (12.3)$$

is A284369 [9]. The parent entry and the zero-position entry A284370 conjecture [10]

$$-3 < n \frac{3 + \sqrt{3}}{2} - u(n) < 4, \quad (12.4)$$

while the parent entry gives the one-position conjecture

$$-2 < n\sqrt{3} - v(n) < 3. \quad (12.5)$$

The standalone A284371 page currently omits the term $-a(n)$ in its printed inequality, but its title, complement relation, and the parent entry make the intended statement (12.5) unambiguous [11]. [Added 1 October 2026, batch 73O1: this describes the A284371 page as the manuscript saw it on 1 October 2026; it was not re-checked at intake.]

12.2 Specific resolution

Theorem 12.1 (Sharp OEIS envelopes). *For the word A284368, put*

$$E_0(n) = n \frac{5 + \sqrt{13}}{2} - u(n), \quad E_1(n) = n \frac{\sqrt{13} - 1}{2} - v(n).$$

Then, for every $n \geq 1$,

$$\frac{1 + \sqrt{13}}{3} < E_0(n) < \frac{4 + \sqrt{13}}{3}, \quad (12.6)$$

$$\frac{\sqrt{13} - 5}{3} < E_1(n) < \frac{\sqrt{13} - 2}{3}. \quad (12.7)$$

The closures of the two error-value sets are the corresponding closed intervals, and every displayed endpoint is a subsequential limit. Consequently both conjectures in (12.2) hold, with substantial room to spare.

For the word A284369, put

$$F_0(n) = n \frac{3 + \sqrt{3}}{2} - u(n), \quad F_1(n) = n\sqrt{3} - v(n).$$

Then, for every $n \geq 1$,

$$-\frac{3 + \sqrt{3}}{2} < F_0(n) < 2 + \sqrt{3}, \quad (12.8)$$

$$-2 < F_1(n) < 1 + \sqrt{3}. \quad (12.9)$$

Again the closures are exactly the corresponding closed intervals, with all four endpoints occurring as subsequential limits. Hence (12.4)–(12.5) are true, and the constants in them can be sharpened to those above.

The A284368 intervals have width exactly one, so they also recover explicit floor formulas.

Corollary 12.2 (The generalized-Wythoff specialization). *For A284368,*

$$v(n) = \left\lfloor n \frac{\sqrt{13} - 1}{2} + \frac{5 - \sqrt{13}}{3} \right\rfloor, \quad (12.10)$$

$$u(n) = \left\lfloor n \frac{5 + \sqrt{13}}{2} - \frac{1 + \sqrt{13}}{3} \right\rfloor. \quad (12.11)$$

These formulas agree with the OEIS identification of the two position sequences as A184484–A184485; no independent priority claim is made for this corollary.

Proof. For example, (12.7) says that $0 < n(\sqrt{13} - 1)/2 + (5 - \sqrt{13})/3 - v(n) < 1$, which is equivalent to (12.10). The zero formula follows identically from (12.6). $\square$

The decimal endpoints are summarized in Table II.2.

Table II.2: Conjectured and sharp position-error envelopes.

word	letter	OEIS interval	exact sharp interval
A284368	0	(1, 3)	(1.5351837584..., 2.5351837584...)
A284368	1	(−1, 1)	(−0.4648162415..., 0.5351837584...)
A284369	0	(−3, 4)	(−2.3660254037..., 3.7320508075...)
A284369	1	(−2, 3)	(−2, 2.7320508075...)

The proof comes from a two-parameter theorem. It is close in spirit to the state-dependent contraction used in Part I for $1 \mapsto (10)^m$ (Sections 3–4), but the internal block geometry is different and produces a distinct Bellman system.

13 The two-parameter family

Fix integers $a, b \geq 1$ and define

$$\tau_{a,b}(0) = 1, \quad \tau_{a,b}(1) = 10^a 1^b. \quad (13.1)$$

Because the image of 1 begins with 1, the iterates $\tau_{a,b}^k(1)$ are nested prefixes of a fixed infinite word $\mathbf{w}^{(a,b)}$. The substitution is primitive. The relation between subdominant substitution eigenvalues and discrepancy is classical [5, 4]; the point here is to solve the legal-prefix state system exactly for this family.

Its incidence matrix, with rows and columns ordered 0, 1, is

$$M = \begin{pmatrix} 0 & a \\ 1 & b + 1 \end{pmatrix}. \quad (13.2)$$

Set

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2}, \quad \lambda = b + 1 + q = \frac{a}{q}. \quad (13.3)$$

Then

$$q^2 + (b+1)q = a, \quad (13.4)$$

and the eigenvalues of M are λ and $-q$. A positive right Perron eigenvector is $(q, 1)^T$, so the zero and one frequencies are

$$\delta = \frac{q}{1+q}, \quad \rho = \frac{1}{1+q}. \quad (13.5)$$

The corresponding position slopes are

$$r = \frac{1}{\delta} = 1 + \frac{1}{q}, \quad s = \frac{1}{\rho} = 1 + q. \quad (13.6)$$

The simple inequality

$$q < 1 \iff \sqrt{(b+1)^2 + 4a} < b+3 \iff a < b+2$$

gives, for integer parameters,

$$q < 1 \iff a \leq b+1. \quad (13.7)$$

This is exactly the bounded-discrepancy regime.

Theorem 13.1 (General sharp envelope theorem). *Assume $a, b \geq 1$ and $a \leq b+1$. Define*

$$\widehat{C} = \frac{a-q}{1-q^2}. \quad (13.8)$$

Let $u(n)$ and $v(n)$ be the positions of the n th zero and one in $\mathbf{w}^{(a,b)}$, and let

$$E_0(n) = \left(1 + \frac{1}{q}\right)n - u(n), \quad E_1(n) = (1+q)n - v(n).$$

Then for every $n \geq 1$,

$$\frac{1}{q} - \widehat{C} - 1 < E_0(n) < \frac{C}{q}, \quad (13.9)$$

$$q - \widehat{C} < E_1(n) < q(1 + \widehat{C}). \quad (13.10)$$

Each endpoint is sharp and is approached by an explicit infinite subsequence. No endpoint is attained at a finite index.

The centered one-count discrepancy satisfies, for every prefix length $N \geq 0$,

$$-\frac{\widehat{C}}{1+q} < O(N) - \frac{N}{1+q} < \frac{q(\widehat{C}+1)}{1+q}, \quad (13.11)$$

where $O(N)$ counts ones in the first N letters. Both constants are sharp. In particular the densities in (13.5) exist.

Theorem 13.2 (Spectral phase transition). *For the family (13.1), the prefix discrepancy is bounded if and only if $a \leq b+1$. If $a = b+2$, there are prefixes whose discrepancy grows linearly with substitution depth, hence at least logarithmically with prefix length. If $a \geq b+3$, there are prefixes whose discrepancy grows geometrically with substitution depth.*

Sections 14–15 prove these theorems. The specific interval attractors needed for Theorem 12.1 are resolved in Section 17.

14 Contracting weights and exact prefix states

For a finite binary word P , define the quadratic weight

$$X(P) = |P|_0 - q|P|_1. \quad (14.1)$$

It is additive under concatenation.

Lemma 14.1 (Weight eigenrelation). *For every finite word P ,*

$$X(\tau_{a,b}(P)) = -qX(P). \quad (14.2)$$

Proof. For the letter 0, the two sides both equal $-q$. For the letter 1,

$$X(10^a1^b) = a - q(b+1) = q^2 = -qX(1),$$

using (13.4). Additivity proves the identity for any word. $\square$

A prefix must remember the letter immediately following it. Let

$$S_i = \{X(P) : P \text{ is a prefix of } \mathbf{w}^{(a,b)} \text{ and the next letter is } i\}, \quad i \in \{0, 1\}, \quad (14.3)$$

and let $K_i = \text{cl } S_i$.

Lemma 14.2 (Graph-directed state equations). *If $q < 1$, the compact sets K_0, K_1 are the unique nonempty compact solution of*

$$K_0 = \bigcup_{j=0}^{a-1} (-qK_1 + j - q), \quad (14.4)$$

$$K_1 = (-qK_0) \cup (-qK_1) \cup \bigcup_{\ell=1}^b (-qK_1 + a - \ell q). \quad (14.5)$$

Proof. Let P be a parent prefix and suppose the next parent letter is 1. Inside its image 10^a1^b , the prefixes immediately before the zeros are

$$1, 10, \dots, 10^{a-1},$$

with weights $j - q$ for $0 \leq j < a$. The prefixes immediately before the initial one and the b trailing ones have weights

$$0, \quad a - q, \quad a - 2q, \quad \dots, \quad a - bq.$$

If the next parent letter is 0, its image is the single letter 1, so the only transition is to state 1 with internal weight 0. Combining these internal weights with Lemma 14.1 gives (14.4)–(14.5).

Conversely, every position lies in one of these image blocks, so no transitions are missing. Since all affine maps have contraction ratio $q < 1$, the graph-directed Hutchinson operator is a contraction in the maximum Hausdorff metric. It therefore has a unique compact fixed pair, and the closures of the actual prefix-state sets form that pair. $\square$

The relation

$$a - bq = q(q+1) > 0 \quad (14.6)$$

follows from (13.4). Thus the smallest state-1 internal digit is 0 and the largest is $a - q$. This observation makes the extrema solvable in closed form even when the attractors themselves have gaps.

15 Solving the Bellman extremum system

Write

$$\ell_i = \min K_i, \quad u_i = \max K_i.$$

Because multiplication by $-q$ reverses order, equations (14.4)–(14.5) give

$$\ell_0 = -qu_1 - q, \quad u_0 = -q\ell_1 + a - 1 - q, \quad (15.1)$$

$$\ell_1 = \min\{-qu_0, -qu_1\}, \quad u_1 = \max\{-q\ell_0, -q\ell_1 + a - q\}. \quad (15.2)$$

Proposition 15.1 (Exact state extrema). *In the bounded regime $q < 1$, with $\widehat{C}$ as in (13.8),*

$$\boxed{\ell_0 = -q(\widehat{C} + 1), \quad u_0 = \widehat{C} - 1, \quad \ell_1 = -qC, \quad u_1 = \widehat{C}.} \quad (15.3)$$

None of these four values belongs to the corresponding unclosed state set S_i , but each is a limit of values in that state.

Proof. The extremum map defined by the right sides of (15.1)–(15.2) is q -Lipschitz in the supremum norm, because minima and maxima are nonexpansive. It therefore has at most one fixed point.

We verify the displayed candidate. Since $u_1 = \widehat{C} > \widehat{C} - 1 = u_0$,

$$\ell_1 = -qu_1 = -qC.$$

Also

$$-q\ell_1 + a - q = q^2\widehat{C} + a - q = \widehat{C}$$

by $\widehat{C}(1 - q^2) = a - q$. The competing maximum branch is

$$-q\ell_0 = q^2(\widehat{C} + 1) < \widehat{C},$$

because

$$\widehat{C} - q^2(\widehat{C} + 1) = a - q - q^2 = qb > 0.$$

Thus $u_1 = \widehat{C}$. The formulas for ℓ_0 and u_0 now follow directly; for the latter,

$$-q\ell_1 + a - 1 - q = q^2\widehat{C} + a - q - 1 = \widehat{C} - 1.$$

Uniqueness proves (15.3).

For nonattainment, suppose first that an actual state-1 prefix had weight $\widehat{C}$. The strict inequality between the two maximum branches forces its parent to have state-1 weight $-qC$ and to use the digit $a - q$. Attaining $-qC$ in turn forces a parent of state 1 and weight $\widehat{C}$, using digit 0. Two desubstitutions therefore return to the same extremal requirement while strictly shortening a nonempty prefix, an infinite-descent contradiction. The same argument starts from $-qC$. Attainment of either state-0 endpoint would force attainment of the corresponding state-1 endpoint through (14.4). Finally, iterating the two extremizing transitions from the empty prefix constructs sequences converging to the endpoints, so they are sharp. $\square$

15.1 Counting discrepancy

If P has length N , then

$$X(P) = N - (1 + q)O(N), \quad O(N) - \frac{N}{1 + q} = -\frac{X(P)}{1 + q}. \quad (15.4)$$

Across both states, Proposition 15.1 gives

$$-q(\widehat{C} + 1) < X(P) < \widehat{C},$$

which is exactly (13.11).

15.2 Position errors

Suppose a prefix P is immediately followed by the n th one. Then $|P|_1 = n - 1$ and $|P| = v(n) - 1$, so

$$E_1(n) = (1 + q)n - v(n) = q - X(P). \quad (15.5)$$

Because $X(P) \in S_1$, Proposition 15.1 yields

$$q - \widehat{C} < E_1(n) < q + qC = q(1 + \widehat{C}).$$

Similarly, immediately before the n th zero one has $|P|_0 = n - 1$ and $|P| = u(n) - 1$, whence

$$E_0(n) = \left(1 + \frac{1}{q}\right)n - u(n) = \frac{X(P) + 1}{q}. \quad (15.6)$$

Using $X(P) \in S_0$ gives

$$\frac{1}{q} - \widehat{C} - 1 < E_0(n) < \frac{C}{q}.$$

This proves Theorem 13.1.

16 The bounded-discrepancy phase transition

The preceding contraction proof establishes bounded discrepancy for $q < 1$. To prove that this condition is necessary, one explicit state-1 cycle is enough.

Let P_0 be the empty prefix and recursively define

$$P_{k+1} = \tau_{a,b}^2(P_k) 1 0^a. \quad (16.1)$$

Every P_k is a prefix immediately before a one. If $x_k = X(P_k)$, then

$$x_0 = 0, \quad x_{k+1} = q^2 x_k + a - q. \quad (16.2)$$

The two steps are: use the zero internal digit before the initial one, then use the largest state-1 digit $a - q$ before the first trailing one.

When $q < 1$,

$$x_k = \widehat{C}(1 - q^{2k}) \longrightarrow \widehat{C}, \quad (16.3)$$

which also gives an explicit sharpness sequence. When $q = 1$,

$$x_k = k(a - 1) = k(b + 1),$$

so discrepancy is unbounded linearly in depth. When $q > 1$, the solution of (16.2) grows geometrically like q^{2k} . Together with (13.7), this proves Theorem 13.2.

At the boundary $q = 1$, substitution lengths grow exponentially in k while x_k grows linearly, giving a logarithmic lower bound in the physical prefix length. A matching $O(\log N)$ upper bound follows by desubstituting any prefix: every twofold image has length at least 3, each internal correction is bounded, and the multiplier has absolute value one. For $q > 1$, the constructed prefixes satisfy $|P_k| \asymp \lambda^{2k}$ and $X(P_k) \asymp q^{2k}$, hence exhibit the power law $X(P_k) \asymp |P_k|^{\log_\lambda q}$. A full normalized limit theory is left for future work. These refinements are not needed for the OEIS applications. [Added 6 October 2026, batch 114: Part III supplies this limit theory. At $a = b + 2$ the cluster set of $X(N)/\log N$ is $[-(a-1)/(2 \log a), (a-1)/(2 \log a)]$ (Theorem 32.1), so the $O(\log N)$ bound above holds with the sharp constant: $|X(N)| \leq \frac{a-1}{2 \log a} \log N + O(1)$; and P_k is the first prefix of weight $k(a-1)$, with an exact length formula (Theorem 32.3). For $a \geq b+3$ the cluster set of $X(N)/N^\alpha$, $\alpha = \log_\lambda q$, is $[-C_{a,b}, C_{a,b}]$ with $C_{a,b}$ in closed form (Theorem 33.1), and $X(P_k)/|P_k|^\alpha \rightarrow C_{a,b}$ (Remark 23.1).]

17 When the state attractors are full intervals

Proposition 15.1 determines the hulls of K_0, K_1 for every bounded parameter pair. It does not assert that the compact sets fill their hulls. Indeed, for some larger parameters the first-level images already leave gaps. The two OEIS cases are especially tractable: one is an exact tiling and the other has ample overlap.

17.1 A284368: an exact interval tiling

Here $(a, b) = (1, 2)$,

$$q = \frac{\sqrt{13} - 3}{2}, \quad \widehat{C} = \frac{\sqrt{13} + 1}{6}. \quad (17.1)$$

Proposition 15.1 gives the candidate intervals

$$I_0 = \left[\frac{2 - \sqrt{13}}{3}, \frac{\sqrt{13} - 5}{6} \right], \quad (17.2)$$

$$I_1 = \left[\frac{\sqrt{13} - 5}{6}, \frac{\sqrt{13} + 1}{6} \right]. \quad (17.3)$$

The state-0 equation has only one image, $I_0 = -qI_1 - q$. The four state-1 images tile I_1 exactly, in the order shown in Table II.3.

Table II.3: Exact interval tiling for $(a, b) = (1, 2)$. Consecutive endpoints coincide.

source and digit	left endpoint	right endpoint
$-qI_1$	$(\sqrt{13} - 5)/6$	$-7/3 + 2\sqrt{13}/3$
$-qI_0$	$-7/3 + 2\sqrt{13}/3$	$19/6 - 5\sqrt{13}/6$
$-qI_1 + 1 - 2q$	$19/6 - 5\sqrt{13}/6$	$5/3 - \sqrt{13}/3$
$-qI_1 + 1 - q$	$5/3 - \sqrt{13}/3$	$(\sqrt{13} + 1)/6$

Thus (I_0, I_1) satisfies the graph-directed equations as sets. Uniqueness in Lemma 14.2 proves

$$K_0 = I_0, \quad K_1 = I_1. \quad (17.4)$$

Applying the affine error maps (15.5)–(15.6) gives exactly (12.6)–(12.7) and proves the A284368 half of Theorem 12.1.

17.2 A284369: overlapping interval covers

Here $(a, b) = (2, 1)$,

$$q = \sqrt{3} - 1, \quad \widehat{C} = 1 + \sqrt{3}. \quad (17.5)$$

The state hulls are

$$I_0 = [-1 - \sqrt{3}, \sqrt{3}], \quad I_1 = [-2, 1 + \sqrt{3}]. \quad (17.6)$$

For state 0, the two image intervals are

$$[-1 - \sqrt{3}, \sqrt{3} - 1], \quad [-\sqrt{3}, \sqrt{3}], \quad (17.7)$$

which overlap and cover I_0 . For state 1, the three images are

$$[-2, 2\sqrt{3} - 2], \quad [\sqrt{3} - 3, 2], \quad [1 - \sqrt{3}, 1 + \sqrt{3}]. \quad (17.8)$$

They overlap consecutively and cover I_1 . Therefore the candidate pair is again an exact set-valued fixed point, and uniqueness gives

$$K_0 = I_0, \quad K_1 = I_1. \quad (17.9)$$

The position-error maps yield (12.8)–(12.9). This completes the proof of Theorem 12.1.

Remark 17.1. The interval property is not automatic. For example, at $(a, b) = (2, 5)$ the two first-level state-0 images have a gap, although the extremal formulas of Proposition 15.1 remain valid. Classifying exactly when all gaps close is one of the main open directions in Section 21.

18 Constructive extremal subsequences and generating functions

The cycle (16.1) not only proves sharpness but gives explicit indices. Let

$$c_k = \begin{pmatrix} z_k \\ o_k \end{pmatrix} = \begin{pmatrix} |P_k|_0 \\ |P_k|_1 \end{pmatrix}. \quad (18.1)$$

Since the suffix 10^a has count vector $(a, 1)^T$,

$$c_0 = 0, \quad c_{k+1} = M^2 c_k + \begin{pmatrix} a \\ 1 \end{pmatrix}. \quad (18.2)$$

The next one has rank $o_k + 1$ and position $z_k + o_k + 1$. Its error is $q - x_k$, which decreases to $q - \widehat{C}$. To exhibit all four endpoints explicitly, put

$$Q_k = \tau_{a,b}(P_k), \quad R_k = \tau_{a,b}(P_k)1, \quad T_k = \tau_{a,b}(Q_k)10^{a-1}.$$

Then Q_k is in state 1 with weight $-qx_k \rightarrow -qC$, so its one-position errors tend to $q(1 + \widehat{C})$. The prefixes R_k and T_k are in state 0 and have respective weights

$$-qx_k - q \rightarrow -q(\widehat{C} + 1), \quad -qX(Q_k) + a - 1 - q \rightarrow \widehat{C} - 1.$$

Their zero-position errors therefore tend to $1/q - \widehat{C} - 1$ and $\widehat{C}/q$. This supplies concrete infinite subsequences for every endpoint in Theorem 13.1.

The vector generating function is rational:

$$\sum_{k \geq 0} c_k z^k = \frac{z}{1 - z} (I - zM^2)^{-1} \begin{pmatrix} a \\ 1 \end{pmatrix}. \quad (18.3)$$

Let $N_k = z_k + o_k = |P_k|$. In the two OEIS cases,

$$\sum_{k \geq 0} N_k z^k = \frac{z(2 - 5z)}{(1 - z)(1 - 11z + z^2)} \quad \text{for A284368,} \quad (18.4)$$

$$\sum_{k \geq 0} N_k z^k = \frac{3z(1 - 2z)}{(1 - z)(1 - 8z + 4z^2)} \quad \text{for A284369.} \quad (18.5)$$

The first terms are

A284368: 0, 2, 19, 204, 2222, 24235, 264360, 2883722, ... ,

A284369: 0, 3, 21, 153, 1137, 8481, 63297, 472449,

Thus the corresponding homogeneous recurrences are

$$N_k = 12N_{k-1} - 12N_{k-2} + N_{k-3} \quad \text{for A284368,}$$

$$N_k = 9N_{k-1} - 12N_{k-2} + 4N_{k-3} \quad \text{for A284369.}$$

These formulas make the sharpness sequences reproducible without building the full fixed words.

19 Computational audit

The verifier code/`02-morphic-verify.py` uses only the Python standard library. It represents every algebraic quantity as $A + B\sqrt{D}$ with rational coefficients, compares signs by integer arithmetic, and never uses a decimal to decide an inequality. Its main checks are:

1. the general Bellman solution for all 119 parameter pairs $1 \leq b \leq 14, 1 \leq a \leq b + 1$;
2. exact interval-tiling and overlap identities for the two OEIS cases;
3. literal generation and comparison with the posted initial OEIS terms;
4. exact checking of every zero- and one-position error in the first 100 000 letters of each word;
5. the extremal vector recurrences and data tables used above.

The recorded run (`data/02-morphic-verification.json`) reports 201 686 exact or counting checks. This finite audit is not used in place of the proofs; it is intended to catch indexing, letter-order, and algebraic-simplification errors.

Figures 2 and 3 plot the first 180 000 letters. The dashed lines are the exact endpoints, not fitted values. The apparent filling of the vertical bands is explained by the interval-attractor proofs in Section 17.

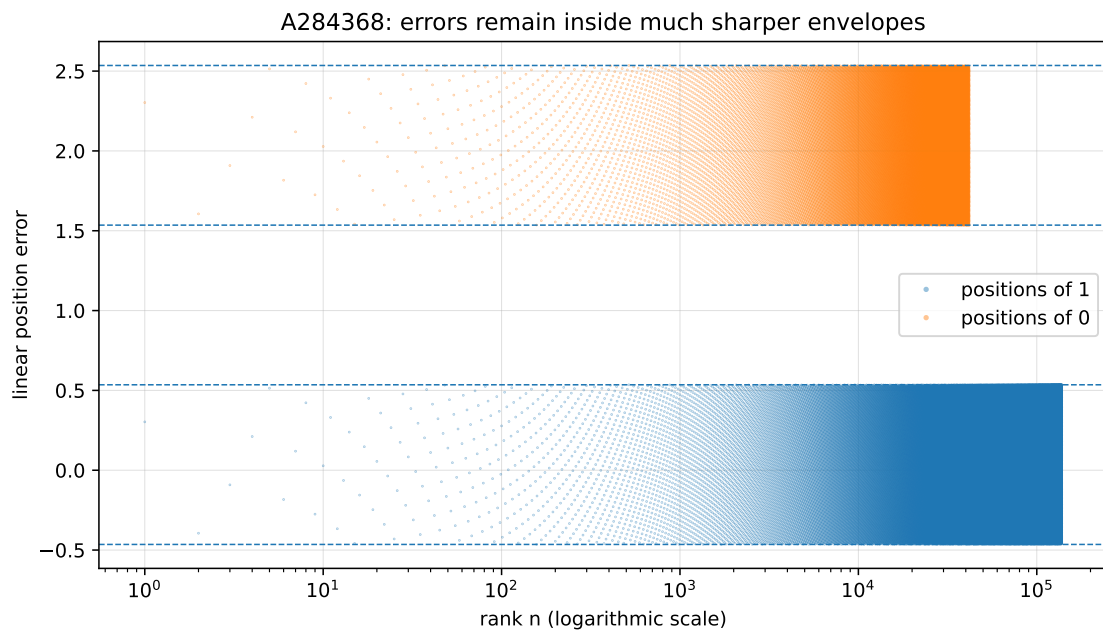


Figure 2: Position errors for A284368. The zero and one errors lie in two unit-width sharp intervals; the displayed envelope constants are exact.

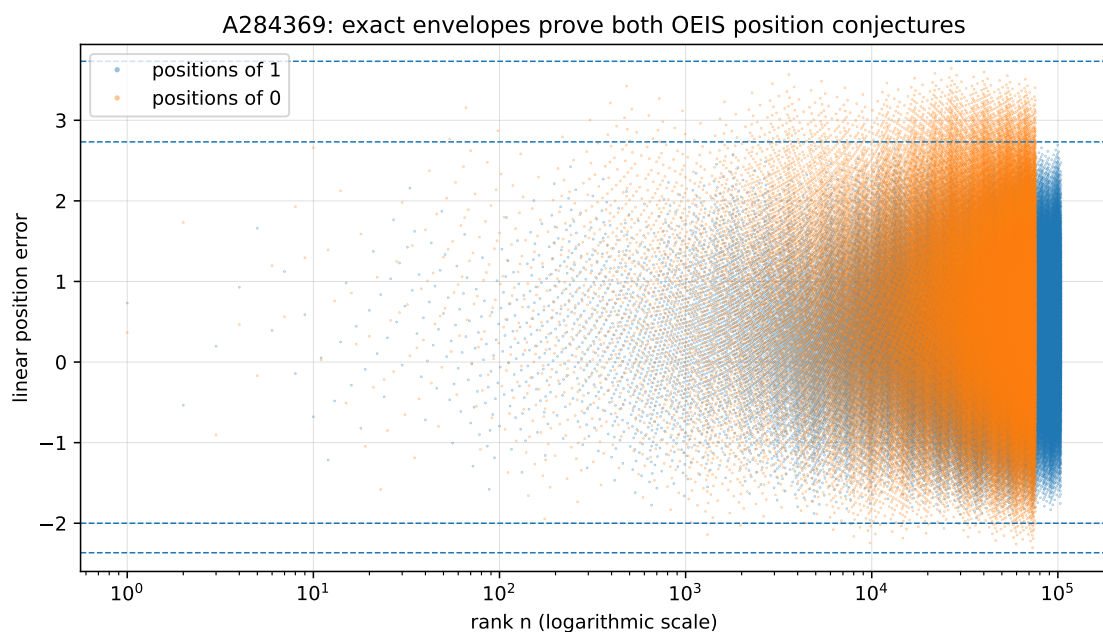


Figure 3: Position errors for A284369. The lower one-error endpoint is exactly -2 but is never attained, proving the strict OEIS inequality.

20 OEIS-ready consequences

The following concise statements can be adapted into OEIS comments after independent review.

A284368 / A184484 / A184485. For the fixed point of $0 \mapsto 1, 1 \mapsto 1011$, let $u(n)$ and $v(n)$ be the positions of the n th zero and one. Then

$$\frac{1 + \sqrt{13}}{3} < n \frac{5 + \sqrt{13}}{2} - u(n) < \frac{4 + \sqrt{13}}{3},$$

$$\frac{\sqrt{13} - 5}{3} < n \frac{\sqrt{13} - 1}{2} - v(n) < \frac{\sqrt{13} - 2}{3}.$$

All four constants are sharp but unattained. The zero and one densities are $(5 - \sqrt{13})/6$ and $(1 + \sqrt{13})/6$.

A284369 / A284370 / A284371. For the fixed point of $0 \mapsto 1, 1 \mapsto 1001$, let $u(n)$ and $v(n)$ be the positions of the n th zero and one. Then

$$-\frac{3 + \sqrt{3}}{2} < n \frac{3 + \sqrt{3}}{2} - u(n) < 2 + \sqrt{3},$$

$$-2 < n \sqrt{3} - v(n) < 1 + \sqrt{3}.$$

All four constants are sharp but unattained. The zero and one densities are $1 - 1/\sqrt{3}$ and $1/\sqrt{3}$.

No OEIS edit has been submitted as part of this work.

21 Further research questions

The two-state reduction exposes several problems that are both concrete and substantially broader than the four resolved inequalities.

1. **Interval-versus-Cantor classification.** Determine necessary and sufficient conditions on (a, b) , with $a \leq b + 1$, for K_0 and K_1 to equal their convex hulls. First-level overlap gives explicit inequalities, but deeper exact overlaps can close apparent gaps. A finite-state classification may exist because all translations lie in the same quadratic field.
2. **Invariant error distributions.** The present paper identifies the support and its endpoints. Determine the limiting empirical measures of $E_0(n)$ and $E_1(n)$, their continuity or singularity, moments, and local dimensions. In interval cases, decide whether the measures are absolutely continuous or have self-similar density singularities. [Added 6 October 2026, batch 114: for the critical family $a = b + 2$, and for every word with a zeros and $a - 1$ ones beginning with 1, Part III determines the law of the prefix weight under uniform prefix sampling: exact complete-block histograms (Theorem 26.2), Gaussian and local limit laws at every cutoff (Theorems 28.1 and 28.3) and large deviations (Theorem 29.2). The bounded regime $a \leq b + 1$ asked about here remains open (Question 1 of Section 35).]
3. **Threshold densities and record statistics.** For a fixed threshold t , determine the natural density of ranks with $E_i(n) > t$, when it exists. This directly continues an open direction

in Part I (A284364–A284366; Section 9), where sharp supports are known but violation densities are not. [Added 6 October 2026, batch 114: at $a = b + 2$, where the errors are unbounded, (29.9) gives the frequency of prefixes with $X(n) \geq x \log_a N$ up to a factor $N^{o(1)}$, namely $N^{-I_a(x)/\log a + o(1)}$. The densities asked for here, in the bounded regime, remain open.]

4. **Exact first-hitting algorithms.** Develop automata or Ostrowski-type numeration systems that return the first rank at which a prescribed algebraic error interval is entered. The graph system suggests a branch-and-bound algorithm with exact quadratic endpoints. [Added 6 October 2026, batch 114: at $a = b + 2$, Theorem 32.3 gives the first prefixes reaching the extremal weights $k(a - 1)$ and $-k(a - 1) - 1$, with closed lengths and ranks. First hits of general prescribed intervals, and the bounded regime, remain open.]
5. **Closed forms for A284370 and A284371.** A284368 is linked to generalized Wythoff sequences. Determine whether the A284369 position sequences admit finite generalized-Beatty formulas, nested floor formulas, or low-complexity transducers. The broad interval support makes a single ordinary Beatty formula impossible, but structured finite combinations may remain possible.
6. **Factor balance rather than prefix balance.** Compute the optimal constant controlling the difference in letter counts between arbitrary equal-length factors. Prefix envelopes give immediate coarse bounds, but the sharp factor constant requires the difference set $K_i - K_j$ together with legal endpoint states.
7. **Critical and supercritical normalization.** At $a = b + 2$, determine the exact limsup and liminf after division by $\log N$. For $a \geq b + 3$, normalize by the predicted power $N^{\log_\lambda q}$ and identify the resulting graph-directed limit sets. [Added 6 October 2026, batch 114: answered in Part III. At $a = b + 2$, $\limsup X(N)/\log N = -\liminf X(N)/\log N = (a - 1)/(2 \log a)$, and every value in between is a limit point (Theorem 32.1). For $a \geq b + 3$, with $\alpha = \log_\lambda q$, the limsup and liminf of $X(N)/N^\alpha$ are $\pm C_{a,b}$, the cluster set is the whole interval, and $X(N) = c^\alpha N^\alpha \Theta(\log_\lambda(cN)) + O(N^{\alpha^2})$ with a continuous anti-periodic profile Θ obtained from a two-state graph-directed system (Theorem 33.1); Corollary 33.2 gives the induced empirical laws. Open: the regularity of Θ and of these laws, and whether the error $O(N^{\alpha^2})$ is sharp (Questions 6 and 7 of Section 35).]
8. **Automated OEIS survey.** Apply the state-equation pipeline to other primitive binary morphisms whose OEIS entries contain unproved Beatty-type error bounds. The steps—incidence spectrum, eigenweight, legal internal digits, Bellman extrema, and overlap audit—are sufficiently explicit to automate candidate discovery while retaining exact proof certificates.
9. **Formalization in Lean.** Formalize finite-word counts, substitution desubstitution, quadratic-field order, and the contracting extremum map in ProveIt. The compact-attractor statement can either be formalized directly or replaced, for the extremal bounds, by finite-depth inequalities plus a geometric tail estimate.

22 Conclusion

The position conjectures for A284369 are not isolated numerical accidents. They are the $(a, b) = (2, 1)$ specialization of a two-parameter theorem whose constants are forced by a four-variable contracting Bellman system. The same theorem gives an independent, sharper resolution of the neighboring A284368 bounds and reveals a clean spectral transition at $a = b + 2$.

The main structural lesson is that the non-Perron eigenvalue gives the contraction rate, but sharp constants require legal prefix states. Once those states are retained, the exact extrema collapse to the single algebraic constant $\widehat{C} = (a - q)/(1 - q^2)$. Whether the resulting compact sets fill their hulls is a separate overlap problem: it succeeds by exact tiling for A284368, by overlap for A284369, and fails already for some nearby parameters. That separation between sharp envelopes and internal fractal geometry provides a natural next research program.

Part III. Exact fluctuation laws and sharp discrepancy in binary morphic words

Positive convolution formulas, extremal variance, and sharp critical and supercritical normalization

Research prepared with ChatGPT, dated 5 October 2026 (the manuscript's title page: "A study of an open question in Vladimir Reshetnikov's ProveIt"); added to this report on 6 October 2026 in batch 114 of ProveIt's incoming-report intake.

Part III answers research question 7 of Part II (Section 21): the exact normalization of the unbounded prefix discrepancy of $0 \mapsto 1, 1 \mapsto 10^a 1^b$ at $a = b + 2$ and for $a \geq b + 3$. Its critical analysis covers every word with a zeros and $a - 1$ ones that begins with 1. Parts I and II are unchanged except for dated notes of 6 October 2026 (Section 16, research questions 2, 3, 4 and 7 of Section 21, and the introduction of Part II) and a front matter that now announces Part III. Section 23 records the provenance, fixes the notation of the addition, and relates it to Parts I and II.

Summary (the manuscript's abstract). The ProveIt study of the substitutions $0 \mapsto 1, 1 \mapsto 10^a 1^b$ leaves the exact normalization of unbounded prefix discrepancy open. We solve its critical case $a = b + 2$: the discrepancy $X(N) = \#0 - \#1$ has the complete logarithmic cluster interval $[-(a - 1)/(2 \log a), (a - 1)/(2 \log a)]$, with explicit first extremal prefixes and exact formulas for their lengths. For $a \geq b + 3$, we also determine the closed supercritical amplitude and the entire normalized cluster interval by an invariant polygon envelope.

The main structural result is broader. For every word w beginning with 1 and containing a zeros and $a - 1$ ones, the complete proper-prefix histogram of $(0 \mapsto 1, 1 \mapsto w)^{2^m}(c)$ equals $1 + K_c(t) \sum_{j < m} [S(t)S(t^{-1})]^j$, with nonnegative integer coefficients throughout. Here S records the heights preceding the zeros of w . This gives an exact mixture of independent digit differences, explicit finite moments, a local Gaussian expansion, quantitative Gaussian laws for every cutoff, and an explicit large-deviation pressure. Among all arrangements with the same incidence matrix, a single consecutive run of zeros maximizes the variance, at $(a^2 - 1)/12$ per substitution depth.

The general central limit theorem and the underlying invariant-section identity have established precedents, which we identify. The contribution is the exact positive representation and the sharp, evaluated consequences. Proofs are accompanied by exact computations and a research agenda.

Status and claim boundary (the manuscript's title page). The manuscript is self-contained at the level of conventional proofs, with standard probability estimates explicitly identified. The computations check finite identities; they are not substitutes for the asymptotic arguments. No proof-assistant verification or external peer review is claimed. [Added 6 October 2026, batch 114: the "local Gaussian expansion" of the summary is proved to leading order (Theorems 28.1 and 28.3); its first correction term (28.9) is sketched only (Question 11).]

23 Provenance, conventions and relation to Parts I and II

Part III is built from one manuscript: *Exact Fluctuation Laws and Sharp Discrepancy in Binary Morphic Words* (5 October 2026; author line "Research prepared with ChatGPT"; 30-page

PDF, 2,187-line source, 117 labels), delivered as `binary_morphic_fluctuations.zip` in arrival commit `e4d5dcf9e` (one of the two “New research reports” commits of 5 October 2026) and placed in commit `99053b5d1` (batch 114). The manuscript read this report at ProveIt commit `112d6bed` (its pin; it also mentions a search-index commit `52d8ca404`, which it did not use). This report’s directory has no difference between `112d6bed` and the placement commit, so Parts I and II as the manuscript read them are Parts I and II here, apart from the dated notes of this write. Its programs, data and figures are shipped with the prefix `03-fluct-` (the README lists them). The merge made these choices.

- Every theorem, corollary, proposition, lemma, definition, remark, proof, table, figure and research item of the manuscript is printed. Its section n is Section $n + 23$ here (its Theorem 3.2, the positive convolution formula, is Theorem 26.2); its four figures are Figures 4–7. Its abstract is the summary above and its title-page note is the status box.
- “The ProveIt report”, “the predecessor” and “the repository” in Sections 24–36 mean Part II of this report as read at `112d6bed`. The manuscript’s bibliography entry for the repository is replaced by references to Part II’s sections; its entries for Adamczewski’s two papers and for A284368–A284371 are merged with the existing ones (the manuscript accessed those OEIS pages on 5 October 2026, the existing entries record 1 October 2026).
- All 117 delivered labels carry the prefix `bmf:`; five labels are new (`bmf:sec:provenance`, `bmf:tab:notation`, `bmf:rem:partII`, `bmf:q:edgeworth`, `bmf:q:fixed-orders`). No symbol is renamed and no normalization changed; Table III.1 resolves every clash with Parts I and II by scope.
- File names in the text are the shipped names, with dated notes in Section 34 where the delivered programs and build script still use delivery names.
- Vladimir’s standing rule of 4 October 2026: two statements that the manuscript argues only in outline (the first local correction (28.9) and every fixed correction order) are marked in place and recorded as Questions 11 and 12 of Section 35. No claim of the manuscript was found to be wrong.

Citations checked at the write (6 October 2026). Solomyak, *On the Lyapunov spectrum of the twisted cocycle for substitutions* (arXiv:2502.14308, version 3 of 20 August 2025): its Proposition 3.1, attributed there to Rajabzadeh and Safaee, states that the vector function $\mathcal{R}(\mathbf{z}) = (1 - z_0, \dots, 1 - z_{d-1})^\top$ is an invariant section of the twisted cocycle, $\mathcal{M}_\zeta(\mathbf{z})\mathcal{R}(\mathbf{z}) = \mathcal{R}(E_\zeta \mathbf{z})$. At $z_0 = t$, $z_1 = t^{-1}$ the row of letter 1 is the identity $(1 - t)S(t) + (1 - t^{-1})T(t) = 1 - t$ used for (26.7) ([write]: writing $\mathbf{z}^p = z_0^{|p|_0} z_1^{|p|_1}$, the row of the letter 1 telescopes, $\sum_{p \in c} \text{a prefix of } w \mathbf{z}^p (1 - z_c) = 1 - \mathbf{z}^w$, and at $z_0 = t$, $z_1 = t^{-1}$ one has $\mathbf{z}^p = t^{X(p)}$ and $X(w) = 1$), so the manuscript’s “one-variable specialization” is accurate. The publisher’s page gives *Studia Math.* **286** (2026), 189–206 (online first 15 December 2025), doi:10.4064/sm250402-31-7; the issue number “no. 2” of the delivered reference was not confirmed. Of Rajabzadeh–Safaee (arXiv:2501.16824) and Bressaud–Bufetov–Hubert (arXiv:1106.2666, *Proc. London Math. Soc.* 2014) only the abstracts were read; the latter’s limit distributions are along exponential subsequences of times for constant-length substitutions, as the manuscript says. Paquette–Son (arXiv:1505.01428, version 1 of 6 May 2015) was read: its Theorem 3.3 is the central limit theorem for $S_f(u_{K_N})/\sqrt{\log N}$ with K_N uniform on $\{1, \dots, N\}$, every N , for a left eigenfunction of modulus-one eigenvalue that is not a coboundary, and its Proposition 3.1(iv)(a) (part of its condition (iv), which is equivalent there to being a coboundary) requires $f(\zeta(a)_j) = \theta_f^{-1}(h(\zeta(a)_j) - h(\zeta(a)_{j+1}))$ at adjacent letters of an image; two adjacent zeros give $f(0) = 0$ ([write]: take $\zeta(a)_j = \zeta(a)_{j+1} = 0$), which fails for the weight $f = (1, -1)$, as the manuscript’s remark after Theorem 28.3 says. The journal

version (Ergodic Theory Dynam. Systems **39** (2019), no. 7, 1971–2005, as cited) was not read; its numbering was not compared. Feller’s Chapter XVI was not checked.

Independent checks. At intake (dossier of batch 114, on copies of the delivered layout): the three standard-library programs and `verify_supercritical.py` pass (37,129 exact assertions; 1,078 words; 1,793 profiles; `all_exact_checks_passed`). Separately programmed spot checks agree with Proposition 32.2 ($a = 3, r \leq 8$; $a = 4, 5, r \leq 6$), with the first-hit lengths (32.6) against literal first occurrences ($a = 3$: 4, 32, 276, 2464 and 9, 89, 817, 7377; $a = 4$: 5, 58, 879 and 12, 215, 3490; $a = 5$: 6, 92, 2178 and 15, 429, 10843), with the profile count (31.1) by brute force over all $\binom{2^a-2}{a}$ words for $a = 2, \dots, 8$, with the slopes $d/2$ of the depth- r extrema for words with $d = 1, 2, 3$, with the exact finite bound $|X(N)| \leq KL_N^\alpha$ for six supercritical pairs $((8, 1), (4, 1), (5, 1), (5, 2), (7, 3), (9, 2))$; 3–12 million prefixes each; largest ratio 0.99976), and with (33.3) on all 2 096 128 prefixes of $\tau^{10}(1)$ at $(a, b) = (8, 1)$, without a violation. At the write: the algebra of the proof of Theorem 27.2 (the seed moments from K_0, K_1 for an arbitrary zero-height law, and the reduction of the mixture variance to (27.8) at both parities) was checked symbolically, and the constants of (28.9) were compared with exact histograms (Question 11).

Notation. Part III’s symbols are local to Sections 23–36. Its critical Sections 25–32 use the weight $X = \#0 - \#1$, which is Part II’s weight $X = \#0 - q\#1$ at $q = 1$; Section 33 uses Part II’s weight itself. Table III.1 prints, for each letter that Parts I–II also use, both meanings. Clashes inside the manuscript, kept as delivered: χ is the pressure and χ_0, χ_1 are the vertical increments of Section 33; d is the diameter of the zero-height support in Sections 25–32 and $d = q + 1$ in Section 33; $e = \lambda - 1$ in Section 33 beside the exponential e^s ; A, B are the length constants (32.5), while A_r, B_r are finite extrema, A is the polygon height of Section 33, $B(t)$ is the one-step matrix and B_j are Bernoulli numbers; $c \in \{0, 1\}$ is a letter, c_h and $c_\pm$ are multiplicities, and $c = (\lambda + q)/(q + 1)$ in Section 33; Y_i are digit differences and $Y(n) = X(n)/n^\alpha$; E is an error constant beside the expectation $\mathbb{E}$; $\log$ is natural and $\log_a$ is to base a .

Remark 23.1 (Relation to Parts I and II). *What is answered.* Part II’s research question 7 asks for the exact limsup and liminf of the discrepancy divided by $\log N$ at $a = b + 2$, and for the limit sets after normalization by $N^{\log_\lambda q}$ at $a \geq b + 3$. The critical family $\tau_{a,a-2}, a \geq 3$, is the run family σ_a (24.3), with $d = a - 1$, so Theorem 32.1 gives the cluster set $[-(a - 1)/(2 \log a), (a - 1)/(2 \log a)]$ of $X(N)/\log N$; this is the sharp constant in the $O(\log N)$ bound of Section 16. In the supercritical range, $\alpha = \log q / \log \lambda$ is Part II’s predicted exponent, Theorem 33.1 gives the cluster set $[-C_{a,b}, C_{a,b}]$ of $X(N)/N^\alpha$, and the graph-directed limit object is the continuous anti-periodic profile Θ , whose range over a period is an interval.

The same cycle [write]. The prefixes P_k of (32.4) and of Section 33 are defined by Part II’s recursion (16.1). At $q = 1$ Part II computes $X(P_k) = k(a - 1)$; Theorem 32.3 adds that P_k is the first prefix of that weight and gives its length. For $q > 1$ the recursion (16.2), $x_{k+1} = q^2 x_k + a - q$, $x_0 = 0$, has the solution $x_k = (a - q)(q^{2k} - 1)/(q^2 - 1)$, which is the weight of P_k printed in Section 33 because $a - q = q(\lambda - 1)$ ($\lambda = a/q$); so Part II’s power law along P_k has the sharp constant, $X(P_k)/|P_k|^\alpha \rightarrow C_{a,b}$. The supercritical $Q_k = \tau(P_k)$ is Part II’s Q_k ; the critical $Q_k = \sigma_a(P_k)1$ of (32.4) is Part II’s Q_k followed by a one.

Position errors [write]. (32.7) is the case $q = 1$ of (15.5) and (15.6): if P immediately precedes the n th one, then $X(v(n)) = X(P) - 1$, so $E_1(n) = -X(v(n)) = 1 - X(P)$; if P immediately precedes the n th zero, then $E_0(n) = X(u(n)) = X(P) + 1$.

Beyond Parts I–II. The critical class of Section 25 contains every word with a zeros and $a - 1$ ones beginning with 1, not only $1 0^a 1^{a-2}$, and it includes $a = 2$ ($w = 100$, that is $(a, b) = (2, 0)$, outside Part II’s hypothesis $b \geq 1$). It is disjoint from Part I’s family: $(10)^m$ has as many ones as zeros. Part III proves nothing about Part I’s words or about Part II’s bounded regime $a \leq b + 1$;

Table III.1: Letters used by Part III and by Parts I–II with different meanings.

Symbol	Parts I–II	Part III
X	$X(P) = P _0 - q P _1$, (3.1), (14.1)	#0 – #1 in Sections 25–32 (Part II's X at $q = 1$); Part II's X in Section 33; $X_a(N)$, $X_q(N)$, X_r , $X(U_N)$ as defined there
σ, W	Part I: $\sigma_m: 1 \mapsto (10)^m$, $W_k = \sigma_m^k(1)$	$\sigma: 1 \mapsto w$ (25.1); $\sigma_a = \tau_{a,a-2}$ (24.3); $W_r = \sigma^r(1)$, $L_r = W_r $
$w, \mathbf{w}$	$\mathbf{w}(\mathbf{w}^{(a,b)})$: the fixed point	w : the finite image $\sigma(1)$ (Sections 25–32), the fixed point of τ (Section 33); $\mathbf{u}$: the fixed point of σ
S	S_0, S_1 : weights of prefixes by next letter	$S(t)$: zero-height polynomial (25.5); $S^-(t) = S(t^{-1})$; $S_j = Y_1 + \dots + Y_j$ (Section 28); S_1, S_2 : two profiles (Question 5)
K_0, K_1	compact sets: Lemma 4.3; $K_i = \text{cl } S_i$	seed polynomials $K_c(t)$, (26.3)–(26.4); K : the amplitude A/t_*^α (Section 33)
C	$C = q/(1 - q^2)$ (2.3); $\widehat{C}$ (13.8)	$C(t)$: the paired matrix (26.9); $C_{a,b}$: the supercritical amplitude
T	$T = m(m + 2)$; T_k : a prefix	$T(t)$: one-height polynomial (25.5); the profile operator (Section 33)
H	Part I's H (2.3)	polygonal paths $H_{i,k}$, H ; Hermite H_2, H_4 ; harmonic H_M ; an accumulator (Section 34)
r, s, t	slopes $r = 1 + 1/q$, $s = 1 + q$; Part I: $t = 1 - q$	r : substitution depth, s : tilt, t : Laurent variable; in Section 33 $r = \lambda^{-2}$, $s = q^{-2}$, t : geometric time
m	Part I's parameter in σ_m	half depth ($r = 2m$); m_r : limiting means; slope $m = e/d$ (Section 33)
u, v	$u(n), v(n)$: positions of the n th zero and one	$v = \text{Var } Z$; $u(n), v(n)$ as in Parts I–II (Section 32); u : a profile variable
ρ, δ	frequencies (13.5)	ρ, ρ_1 : decay rates (Section 28); δ not used
P_k, Q_k	(16.1); $Q_k = \tau_{a,b}(P_k)$	P_k : the same recursion; $Q_k = \sigma_a(P_k)1$ in (32.4), $Q_k = \tau(P_k)$ in Section 33; P : the polygon there
E_0, E_1	position errors (15.5), (15.6)	the same at $q = 1$ (32.7); E : an error constant (Section 33)
I, F, ℓ	I_0, I_1 (4.1); $F_m(z)$, F_0, F_1 ; $\ell_i = \min K_i$	rate functions I_w, I_a , identity I ; histograms $F_{r,c}$ and profile F ; ℓ_0, ℓ_1 : letter lengths, ℓ : least height, an index

Part I's open questions (Section 9) and Part II's research questions other than question 7 stay open (of question 4, Theorem 32.3 answers only the extremal slice at $a = b + 2$) (Part III's own Question 1 says that its critical mixture does not extend across $q = 1$).

Method. Parts I and II solve contracting set recursions and Bellman systems for the extremes. Part III's critical results rest on an exact, coefficientwise positive identity for complete-block prefix histograms (Theorem 26.2), its supercritical result on continuous graph-directed profiles and an invariant concave polygon envelope (Section 33).

What Part III does not claim. Its text keeps every limitation the manuscript states; in summary: no external review or proof-assistant verification; finite computations corroborate the proofs and do not replace them, and the floating-point diagnostics and profile plots are not interval-certified; the spectral explanation of discrepancy growth (Adamczewski), the existence of the Gaussian limit (Paquette–Son, Theorem 3.3) and the invariant-section identity (Rajabzadeh–Safaei, Solomyak) are prior work, and the unnamed “general predecessors” of the continuous profile construction are acknowledged without a claim of a new general limit-profile theorem; the priority statement is a bounded literature assessment; the critical word 10001 is given no OEIS number (A284370 is a position sequence of 1001); the bounded-regime distribution questions are not solved; no absolute continuity or absence of atoms is asserted for the phase laws; Theorem 31.1 counts profiles, not words; the error $O(N^{a^2})$ is not claimed sharp; the endpoint values of the rate function concern neighbourhoods, not exact extremes; and the first local correction is sketched only.

24 The question, the answer, and its provenance

A morphism replaces each letter of a finite word by another finite word and concatenates the replacements. If the image of 1 begins with 1 and grows, its iterates give longer and longer prefixes of one infinite word. Such a word can be completely deterministic while its prefix counts display substantial statistical variation.

The starting point is the ProveIt report *Sharp discrepancy for two families of binary substitutions*, specifically its Part II (Sections 12–21 of this report, read by the manuscript at commit 112d6bed; see Section 23). Its family is (13.1), repeated here:

$$\tau_{a,b}(0) = 1, \quad \tau_{a,b}(1) = 10^a 1^b, \quad a, b \geq 1. \quad (24.1)$$

Writing

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2}, \quad \lambda = b + 1 + q, \quad (24.2)$$

the incidence eigenvalues are λ and $-q$. The predecessor establishes bounded discrepancy for $q < 1$, logarithmic extremal growth at $q = 1$, and power growth for $q > 1$ (Theorem 13.2 and Section 16). Its research question 7 (Section 21) asks for the exact critical limsup and liminf and the supercritical normalized limit sets. Our principal critical results answer the first part exactly and extend it to every arrangement with the same incidence matrix. Section 33 answers the supercritical part as well, including a closed formula for its endpoints.

24.1 The OEIS connection, with the indices kept distinct

The link with OEIS is through the explicitly investigated two-parameter family, rather than an unverified assignment of a new OEIS number to its critical boundary. The entries inspected for this article have the following meanings [6, 9, 10, 11].

Entry	Object	Parameters in (24.1)
A284368	Fixed word of $1 \mapsto 1011$	$(a, b) = (1, 2)$
A284369	Fixed word of $1 \mapsto 1001$	$(a, b) = (2, 1)$
A284370	Positions of zeros in A284369	same bounded case
A284371	Positions of ones in A284369	same bounded case

These examples are in the bounded regime. The critical continuation is

$$\sigma_a(0) = 1, \quad \sigma_a(1) = 10^a 1^{a-2}, \quad a \geq 3. \quad (24.3)$$

For example, $a = 3$ gives $1 \mapsto 10001$. This word is not A284370: A284370 is a position sequence for $1 \mapsto 1001$. The bounded-regime distribution questions in the predecessor remain separate from the critical distribution laws proved below.

24.2 The main concrete conclusions

Let $X_a(N)$ be the number of zeros minus the number of ones among the first N letters of the fixed point of (24.3). We prove

$$\limsup_{N \rightarrow \infty} \frac{X_a(N)}{\log N} = -\liminf_{N \rightarrow \infty} \frac{X_a(N)}{\log N} = \frac{a-1}{2 \log a}. \quad (24.4)$$

Every value between these two endpoints is a subsequential limit. If U_N is uniform on $\{0, \dots, N-1\}$, then

$$\frac{X_a(U_N)}{\sqrt{((a^2-1)/12) \log_a N}} \implies \mathcal{N}(0, 1). \quad (24.5)$$

Thus typical errors are of order $\sqrt{\log N}$, whereas extremal errors are of order $\log N$. These are different assertions about the same deterministic word.

The exact pressure controlling deviations on the larger scale is

$$\chi_a(s) = \log \left(\frac{\sinh(as/2)}{a \sinh(s/2)} \right), \quad \chi_a(0) = 0. \quad (24.6)$$

In particular, the frequency of a prescribed positive logarithmic-size deviation is a power of N with an explicitly determined exponent; see Section 29.

All three conclusions arise from one exact finite identity, rather than extrapolation from sampled prefixes.

In the supercritical range $a \geq b+3$, set $\alpha = \log q / \log \lambda$. For the eigenweight $X_q(N) = \#0 - q\#1$, we obtain the complete cluster interval

$$\text{Clust}_{N \rightarrow \infty} \frac{X_q(N)}{N^\alpha} = [-C_{a,b}, C_{a,b}],$$

with the explicit constant

$$C_{a,b} = \frac{\lambda-1}{q^2-1} \left(\frac{(\lambda+q)(\lambda^2-1)}{(q+1)^2} \right)^\alpha. \quad (24.7)$$

For example, $0 \mapsto 1, 1 \mapsto 10^8 1$ has $q = 2, \lambda = 4, \alpha = 1/2$ and sharp endpoints $\pm\sqrt{10}$.

24.3 What is established theory and what is supplied here

Adamczewski's work supplies the general relation between substitution spectra and bounded or logarithmic discrepancy [5, 4]. Paquette and Son, Theorem 3.3, already prove the existence of a Gaussian limit for noncoboundary eigenfunctions with eigenvalue of modulus one, using a uniformly selected prefix and arbitrary cutoffs [13]. It applies to our class. Our Gaussian limit is therefore an explicit evaluation and refinement of that theorem, with an elementary independent-summand proof.

The telescoping identity used below is a one-variable specialization of the invariant-section identity attributed to Rajabzadeh and Safaee in Solomyak's Proposition 3.1 [14, 15]. We do not claim the general invariant-section mechanism as new. The exact positive histogram mixture, its reduction to the finite zero-height distribution, and the resulting sharp constants are the specific results developed here. We have not located these exact statements in the primary sources surveyed; that is a bounded literature assessment, not a certification of publication priority.

The related work of Bressaud, Bufetov, and Hubert [16] concerns a different sampling framework in its constant-length limit theorems: a random initial point at prescribed times. We keep it distinct from a uniform prefix of one fixed word.

25 A general critical binary class

Fix an integer $a \geq 2$. Let w be any binary word that begins with 1 and has a zeros and $a - 1$ ones. Define

$$\sigma(0) = 1, \quad \sigma(1) = w. \quad (25.1)$$

The substitution is primitive, and its nested iterates on 1 define a fixed point $\mathbf{u} = u_1 u_2 \cdots$. For a finite word P set

$$X(P) = |P|_0 - |P|_1, \quad X(N) = X(u_1 \cdots u_N), \quad X(0) = 0. \quad (25.2)$$

The incidence matrix and its eigenweight identity are

$$M = \begin{pmatrix} 0 & a \\ 1 & a - 1 \end{pmatrix}, \quad X(\sigma(P)) = -X(P). \quad (25.3)$$

The eigenvalues are a and -1 . The eigenweight identity follows already from $X(\sigma(0)) = -1$ and $X(\sigma(1)) = 1$.

For $r \geq 0$ let $W_r = \sigma^r(1)$ and $L_r = |W_r|$. The order of the letters in w does not affect these lengths:

$$L_r = \frac{2a^{r+1} + (-1)^{r+1}(a - 1)}{a + 1}. \quad (25.4)$$

Indeed $L_0 = 1$, $L_1 = 2a - 1$, and $L_{r+1} = (a - 1)L_r + aL_{r-1}$. Its characteristic roots are $a, -1$. For $r \geq 1$, $|\sigma^r(0)| = L_{r-1}$.

Definition 25.1 (Zero-height distribution). For each occurrence of a zero in w , record the value of X immediately before that occurrence. Let Z be a uniformly selected one of these a values, with repetitions retained. Define

$$S(t) = \sum_{\substack{p0 \text{ is a prefix} \\ \text{of } w}} t^{X(p)}, \quad T(t) = \sum_{\substack{p1 \text{ is a prefix} \\ \text{of } w}} t^{X(p)}. \quad (25.5)$$

Thus $S(t)/a$ is the Laurent probability generating function of Z . We write $\mu = \mathbb{E}Z$, $v = \text{Var } Z$, and $d = \max \text{supp } Z - \min \text{supp } Z$.

Lemma 25.2 (The zero heights have no gaps). *The support of Z is a consecutive interval of integers containing -1 and 0 . In particular $v > 0$, $d \geq 1$, and $Z - Z'$ has lattice span one for an independent copy Z' .*

Proof. The prefix weights trace a nearest-neighbor path from 0 to 1 . The first step is downward. The path must later cross upward from -1 to 0 and from 0 to 1 , so both starting heights occur. If a negative height is reached, the subsequent return to the final height 1 includes every intervening upward crossing. If a nonnegative height is reached by an upward crossing, the path from its initial height 0 must include all intervening upward crossings. The set of starting heights of the upward steps is consequently an integer interval. Each such step is exactly a zero occurrence. $\square$

For (24.3), the zero heights are $-1, 0, \dots, a-2$, each once, so

$$\mu = \frac{a-3}{2}, \quad v = \frac{a^2-1}{12}, \quad d = a-1. \quad (25.6)$$

26 An exact positive formula for every prefix histogram

Definition 26.1 (Proper-prefix histogram). For $c \in \{0, 1\}$, let

$$F_{r,c}(t) = \sum_{0 \leq n < |\sigma^r(c)|} t^{X(\text{pref}_n(\sigma^r(c)))}. \quad (26.1)$$

The empty prefix is included and the complete word is excluded. Consequently $F_{0,0} = F_{0,1} = 1$.

Theorem 26.2 (Positive convolution formula). *Let $S^-(t) = S(t^{-1})$ and $T^-(t) = T(t^{-1})$, and put*

$$\Delta(t) = S(t)S^-(t), \quad (26.2)$$

$$K_0(t) = S(t) + T(t) - 1, \quad (26.3)$$

$$K_1(t) = K_0(t^{-1}) + T^-(t)K_0(t). \quad (26.4)$$

Both K_c have nonnegative integer coefficients, and

$$K_0(1) = 2(a-1), \quad K_1(1) = 2a(a-1). \quad (26.5)$$

For every $m \geq 0$ and $c \in \{0, 1\}$,

$$F_{2m,c}(t) = 1 + K_c(t) \sum_{j=0}^{m-1} \Delta(t)^j. \quad (26.6)$$

The sum is empty when $m = 0$. Odd depths for W_r are recovered by $F_{r,1} = F_{r+1,0}$.

Proof. An upward step from height h changes t^h by $(t-1)t^h$; a downward step changes it by $(t^{-1}-1)t^h$. Summing these increments along w , whose final height is 1 , gives the Laurent-polynomial identity

$$t-1 = (t-1)S(t) + (t^{-1}-1)T(t).$$

It follows that

$$T(t) = t(S(t)-1). \quad (26.7)$$

This identity holds formally, including at $t = 1$ by evaluation.

Every proper prefix of $\sigma^{r+1}(c)$ selects one letter in $\sigma(c)$ and then a proper prefix in its σ^r image. The preceding complete images contribute $(-1)^r X(p)$. With column vectors $F_r = (F_{r,0}, F_{r,1})^T$, this gives

$$F_{r+1}(t) = B(t^{(-1)^r})F_r(t), \quad B(t) = \begin{pmatrix} 0 & 1 \\ S(t) & T(t) \end{pmatrix}. \quad (26.8)$$

Pairing two steps therefore gives

$$F_{2m} = C(t)^m \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad C(t) = B(t^{-1})B(t) = \begin{pmatrix} S & T \\ T^{-1}S & S^{-1} + T^{-1}T \end{pmatrix}. \quad (26.9)$$

Equation (26.7) implies $T^{-1}T = (S^{-1} - 1)(S - 1)$, so

$$\text{tr } C = \Delta + 1, \quad \det C = \Delta.$$

The Cayley–Hamilton identity becomes $(C - I)(C - \Delta I) = 0$. Induction, entirely in the Laurent-polynomial ring, now gives

$$C^m = I + (C - I) \sum_{j=0}^{m-1} \Delta^j.$$

Multiplication by $(1, 1)^T$ yields (26.6), because

$$(C - I) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} S + T - 1 \\ S^{-1} - 1 + T^{-1}(S + T) \end{pmatrix} = \begin{pmatrix} K_0 \\ K_1 \end{pmatrix}.$$

Finally w begins with 1, so T includes the constant term of its empty prefix. Subtracting this one contribution leaves $K_0 \geq 0$ coefficientwise. Formula (26.4) then proves $K_1 \geq 0$. Their total masses follow from $S(1) = a$ and $T(1) = a - 1$. The odd-depth assertion is the word identity $\sigma^{r+1}(0) = \sigma^r(1)$. $\square$

Remark 26.3 (Why positivity matters). The factorization of the two eigenvalues by itself would give a spectral formula. Equation (26.6) gives more: it expresses the integer histogram as a sum of positive convolutions. It therefore has an exact probability interpretation with no signed weights, and no spectral remainder has to be guessed or fitted. The telescoping mechanism is related to the general invariant section in [14, Proposition 3.1]; the positivity and probability interpretation here use the specified binary class.

26.1 A complete small example

Take $a = 3$ and $w = 10001$. Then

$$\begin{aligned} S &= t^{-1} + 1 + t, & T &= 1 + t^2, \\ \Delta &= t^{-2} + 2t^{-1} + 3 + 2t + t^2, & K_0 &= t^{-1} + 1 + t + t^2, \\ K_1 &= t^{-3} + 2t^{-2} + 3t^{-1} + 3 + 2t + t^2. \end{aligned} \quad (26.10)$$

The depth-two word is the concatenation $10001|1|1|1|10001$, of length 13. Its proper-prefix weights are

$$0, -1, 0, 1, 2, 1, 0, -1, -2, -3, -2, -1, 0.$$

Their histogram is $1 + K_1$, exactly as the theorem says. The same identity computes the entire histogram at depth 256, even though the corresponding word has on the order of 3^{256} letters.

27 The exact random-sum law and finite moments

Let Z_i, Z'_i be independent copies of Z , and set $Y_i = Z_i - Z'_i$. Then

$$\mathbb{E}t^{Y_i} = \frac{\Delta(t)}{a^2}, \quad \mathbb{E}Y_i = 0, \quad \text{Var } Y_i = 2v. \quad (27.1)$$

The distribution of Y_i is symmetric and has span one.

Corollary 27.1 (Exact geometric mixture). *Fix $m \geq 1$, $c \in \{0, 1\}$, and let $L = |\sigma^{2m}(c)|$. A uniformly chosen proper-prefix weight has the following law: with probability $1/L$ it is 0; with probability $\omega = (L - 1)/L$ it is*

$$V_c + \sum_{i=1}^J Y_i, \quad (27.2)$$

where all variables on the right are independent,

$$\mathbb{E}t^{V_c} = \frac{K_c(t)}{K_c(1)}, \quad \mathbb{P}(J = j) = \frac{a^{2j}}{\sum_{h=0}^{m-1} a^{2h}}, \quad 0 \leq j < m.$$

Proof. Divide each summand of (26.6) by its total mass. The mass of the summand $K_c \Delta^j$ is $K_c(1)a^{2j}$. $\square$

The depth loss $m - 1 - J$ has a truncated geometric distribution. All of its fixed moments are bounded independently of m . In particular, with $R = a^2$,

$$\mathbb{E}J = m - \frac{R}{R-1} + \frac{m}{R^m - 1}. \quad (27.3)$$

The random sum therefore contains essentially one independent digit difference for every two substitution levels.

27.1 Exact moments for every word in the class

The first two moments can be evaluated using only $\mu = \mathbb{E}Z$ and $v = \text{Var } Z$. Define

$$m_r = \begin{cases} -\frac{1}{2} - \frac{\mu}{a-1}, & r \text{ even}, \\ \frac{1}{2} + \frac{a\mu}{a-1}, & r \text{ odd}, \end{cases} \quad (27.4)$$

$$\beta = \frac{v}{a+1} - \frac{a\mu^2}{(a-1)^2} + \frac{1}{4}, \quad \omega_r = 1 - \frac{1}{L_r}, \quad (27.5)$$

$$\eta_r = \begin{cases} ar, & r \text{ even}, \\ r+1, & r \text{ odd}. \end{cases} \quad (27.6)$$

Theorem 27.2 (Finite mean and variance). *For U_r uniform on $\{0, \dots, L_r - 1\}$ and $X_r = X(U_r)$, the following identities hold for every $r \geq 0$:*

$$\mathbb{E}X_r = \omega_r m_r, \quad (27.7)$$

$$\text{Var } X_r = \omega_r(rv + \beta) + \frac{2v\eta_r}{(a+1)L_r} + \omega_r(1 - \omega_r)m_r^2. \quad (27.8)$$

Consequently

$$\mathbb{E}X_r = m_r + O_w(a^{-r}), \quad \text{Var } X_r = rv + \beta + O_w(ra^{-r}). \quad (27.9)$$

The variance correction β is the same in both parity classes.

Proof. Differentiate the probability generating functions at $t = e^s$, so derivatives at $s = 0$ give moments. From (26.7) and (26.3)–(26.4) one obtains

$$\begin{aligned}\mathbb{E}V_0 &= \frac{1}{2} + \frac{a\mu}{a-1}, & \text{Var } V_0 &= \frac{av}{a-1} - \frac{a\mu^2}{(a-1)^2} + \frac{1}{4}, \\ \mathbb{E}V_1 &= -\frac{1}{2} - \frac{\mu}{a-1}, & \text{Var } V_1 &= \frac{(2a-1)v}{a-1} - \frac{a\mu^2}{(a-1)^2} + \frac{1}{4}.\end{aligned}$$

For example $S(e^s)'|_0 = a\mu$ and $S(e^s)''|_0 = a(v + \mu^2)$; substitution in $K_0 = (1+t)S - t - 1$ gives the first row directly. Substitution in $K_1 = K_0^- + T^-K_0$ gives the second.

Conditional on the non-atomic part of (27.2), the mean is $\mathbb{E}V_c$ and the variance is $\text{Var } V_c + 2v\mathbb{E}J$. The zero atom changes these to $\omega\mathbb{E}V_c$ and $\omega(\text{Var } V_c + 2v\mathbb{E}J) + \omega(1-\omega)(\mathbb{E}V_c)^2$. Use (27.3). For even $r = 2m$, use $c = 1$; for odd $r = 2m-1$, use $c = 0$ at depth $2m$. The lengths from (25.4) reduce the resulting expressions to (27.7)–(27.8). For $r = 0$, $\omega_0 = \eta_0 = 0$, so both sides vanish. $\square$

For the critical run family (24.3), these constants are

$$m_{\text{even}} = -\frac{a-2}{a-1}, \quad m_{\text{odd}} = \frac{a-1}{2} - \frac{1}{a-1}, \quad \beta = \frac{7-a}{6} - \frac{1}{(a-1)^2}. \quad (27.10)$$

At $a = 3$, the limiting means are $-1/2$ and $1/2$, $v = 2/3$, and $\beta = 5/12$. At depth two the exact values are $\mathbb{E}X_2 = -6/13$ and $\text{Var } X_2 = 302/169$.

28 Gaussian laws, local expansions, and arbitrary cutoffs

Throughout this section w is fixed. Constants in $O_w(\cdot)$ may depend on that finite word, but never on the cutoff or substitution depth. Write $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$ and let Φ be its distribution function.

28.1 A bounded independent-summand estimate

The exact mixture makes the relevant probability input elementary. For $S_j = Y_1 + \cdots + Y_j$, the usual Berry–Esseen estimate for bounded independent summands gives

$$\sup_x \left| \mathbb{P}\left(\frac{S_j}{\sqrt{2vj}} \leq x\right) - \Phi(x) \right| = O_w(j^{-1/2}). \quad (28.1)$$

This is a standard probability theorem, not a new general estimate; see, for example, [17, Chapter XVI]. The lattice local estimate can be seen directly. The characteristic function is

$$\psi(t) = \mathbb{E}e^{itY_1} = |\mathbb{E}e^{itZ}|^2 = 1 - vt^2 + O_w(t^4).$$

Lemma 25.2 gives $|\psi(t)| < 1$ for $0 < |t| \leq \pi$. On any closed interval separated from 0 the bound is uniform. Fourier inversion, with a small neighborhood of zero separated from its complement, therefore gives

$$\sup_{k \in \mathbb{Z}} \left| \mathbb{P}(S_j = k) - \frac{1}{\sqrt{2vj}} \varphi\left(\frac{k}{\sqrt{2vj}}\right) \right| = O_w(j^{-3/2}). \quad (28.2)$$

For completeness, in the small neighborhood the integrand difference is bounded by $Cjt^4e^{-cjt^2}$; its integral is $O(j^{-3/2})$. The complementary integral is exponentially small. This also explains the absence of a bulk skewness term.

Theorem 28.1 (Full-block normal and local laws). *For X_r as in Theorem 27.2, as $r \rightarrow \infty$,*

$$\sup_x \left| \mathbb{P} \left(\frac{X_r - m_r}{\sqrt{rv}} \leq x \right) - \Phi(x) \right| = O_w(r^{-1/2}), \quad (28.3)$$

$$\sup_{k \in \mathbb{Z}} \left| \mathbb{P}(X_r = k) - \frac{1}{\sqrt{rv}} \varphi \left(\frac{k - m_r}{\sqrt{rv}} \right) \right| = O_w(r^{-3/2}). \quad (28.4)$$

Proof. For (28.3), apply (28.1) conditional on J in Corollary 27.1. The event $J < m/2$ is exponentially small. On its complement, the bounded seed changes a normal distribution function by $O(m^{-1/2})$, and the scale change from J to m is bounded by $C(m - J)/m$. The geometric depth loss has bounded moments. The atom contributes only $O(a^{-r})$. Odd depths use state 0 at the next even depth.

Here are details yielding the sharper centered local estimate. On a complex neighborhood of $s = 0$, choose the analytic logarithm and square root that are real at 0, and put

$$\chi(s) = \frac{1}{2} \log \frac{S(e^s)S(e^{-s})}{a^2}. \quad (28.5)$$

The finite geometric identity implies

$$\mathbb{E}e^{sX_r} = e^{r\chi(s)} g_\epsilon(s)(1 + O_w(\rho^r)), \quad 0 < \rho < 1, \quad \epsilon = r \bmod 2, \quad (28.6)$$

uniformly on a smaller fixed neighborhood, where

$$g_0(s) = \frac{(a+1)K_1(e^s)}{2a(\Delta(e^s) - 1)}, \quad (28.7)$$

$$g_1(s) = \frac{(a+1)K_0(e^s)\sqrt{\Delta(e^s)}}{2a(\Delta(e^s) - 1)}. \quad (28.8)$$

Both are 1 at 0. The moment identities give $(\log g_\epsilon)'(0) = m_r$ and $(\log g_\epsilon)''(0) = \beta$. Also $\chi'(0) = \chi'''(0) = 0$ and $\chi''(0) = v$. Consequently, after removal of the bounded mean, the small-frequency characteristic function differs from $e^{-rv t^2/2}$ by an integrable error bounded by $C(t^2 + rt^4)e^{-crt^2}$, whose integral is $O(r^{-3/2})$. Away from zero, use the original finite geometric sum, so that possible points $\Delta(e^{it}) = 1$ create no denominator problem: its normalized modulus is bounded by $Cr\rho_1^r$ for some $\rho_1 < 1$. Fourier inversion proves (28.4). $\square$

28.2 The first local correction

Let $\kappa_4(Z) = \mathbb{E}(Z - \mu)^4 - 3v^2$ and set $H_2(y) = y^2 - 1$, $H_4(y) = y^4 - 6y^2 + 3$. Keeping the next terms in the preceding Fourier proof gives, uniformly for $y = (k - m_r)/\sqrt{rv}$ in any fixed compact interval,

$$\mathbb{P}(X_r = k) = \frac{\varphi(y)}{\sqrt{rv}} \left[1 + \frac{1}{r} \left\{ \frac{\beta}{2v} H_2(y) + \frac{\kappa_4(Z)}{24v^2} H_4(y) \right\} + O_w(r^{-3/2}) \right]. \quad (28.9)$$

Indeed $\chi^{(4)}(0) = \kappa_4(Z)$; the second derivative of $\log g_\epsilon$ supplies the H_2 term. Higher odd derivatives of χ vanish. The bounded third derivative of $\log g_\epsilon$ first enters at order $r^{-3/2}$ in the bracket. The same analytic expansion gives every fixed local correction order. This statement concerns fixed orders and does not assert convergence of an infinite asymptotic expansion. [Added 6 October 2026, batch 114: the manuscript argues (28.9) and the preceding sentence on every fixed correction order only in outline (“keeping the next terms”); neither is proved here as a theorem. Both are recorded as Questions 11 and 12 of Section 35, with what is missing. The

constants in (28.9) and (28.10) agree with an exact-histogram check made at the write (same section).]

In the critical run family, $\kappa_4(Z) = -(a^4 - 1)/120$. The displayed correction polynomial is

$$\frac{6\beta}{a^2 - 1}H_2(y) - \frac{a^2 + 1}{20(a^2 - 1)}H_4(y). \quad (28.10)$$

28.3 Why every cutoff has the same leading law

A theorem about L_r alone does not automatically imply a theorem about all N . The required passage uses a precise prefix decomposition.

Lemma 28.2 (Decomposition with a geometric loss of depth). *Let h be the least integer with $N \leq L_h$. The prefix of length N can be partitioned into complete blocks $\sigma^r(c)$ with at most $2a - 2$ blocks at each level $r < h$, or into the single block W_h if it is complete. For a block at level $r = h - j$, its mass under uniform sampling of the N prefix positions is $O_w(a^{-j})$, summed over all blocks at that level. Its starting weight has absolute value $O_w(j + 1)$.*

Proof. Expand the one unfinished block into the images of its letters. Accept the complete children and continue only inside the next unfinished child. There are at most $2a - 2$ accepted children at a stage. By (25.4), a level- r child has length comparable to a^r , while N is comparable to a^h . The mass estimate follows. At an ancestor level, every preceding complete sibling has total weight 1 or -1 by (25.3); a bounded number of siblings is passed at each of the $j + 1$ levels. Their sum is the stated starting weight. $\square$

Theorem 28.3 (Quantitative normal laws at every cutoff). *Let $N \geq 2$, let U_N be uniform on $\{0, \dots, N - 1\}$, and put $b_N = \log_a N$. As $N \rightarrow \infty$,*

$$\mathbb{E}X(U_N) = O_w(1), \quad \text{Var } X(U_N) = vb_N + O_w(1), \quad (28.11)$$

$$\sup_x \left| \mathbb{P} \left(\frac{X(U_N)}{\sqrt{vb_N}} \leq x \right) - \Phi(x) \right| = O_w(b_N^{-1/2}), \quad (28.12)$$

$$\sup_{k \in \mathbb{Z}} \left| \sqrt{vb_N} \mathbb{P}(X(U_N) = k) - \varphi \left(\frac{k}{\sqrt{vb_N}} \right) \right| = O_w(b_N^{-1/2}). \quad (28.13)$$

These conclusions also hold for the one-based convention $U_N \in \{1, \dots, N\}$.

Proof. Condition on the block of Lemma 28.2. Its internal mean is bounded and its variance is $vr + O_w(1)$. The lost depth j and the starting weight have uniformly bounded moments under the geometric mass bound. Conditional expectation and variance therefore give (28.11), using $h = b_N + O_w(1)$.

For the normal estimates first retain $j \leq h/2$. There r is comparable to b_N . The full-block error is $O_w(b_N^{-1/2})$. The bounded seed and starting shift change the normal comparison by at most $C_w(j + 1)/\sqrt{b_N}$, and the variance-scale change contributes at most $C_w(j + 1)/b_N$. Summation over the geometric level masses proves the asserted errors. The discarded mass is $O_w(a^{-h/2})$; for the local assertion multiply it by $\sqrt{b_N}$ before discarding it. This is still negligible. Replacing the sampled interval $\{0, \dots, N - 1\}$ by $\{1, \dots, N\}$ changes the law in total variation by at most $1/N$. Since $X(N) = O_w(\log N)$, as also proved in Section 32, the corresponding changes in the first and second moments are $O_w(\log N/N)$ and $O_w((\log N)^2/N)$. Thus the moment conclusions transfer too. $\square$

Remark 28.4 (Relation to the existing central limit theorem). The existence of the limit in (28.12) is already implied by Paquette–Son [13, Theorem 3.3]. In this class their noncoboundary condition can also be checked directly: w must contain 00, since its first letter is 1 and there are more zeros than ones. Their Proposition 3.1 applied to adjacent equal letters excludes a coboundary for the weight $(1, -1)$. Theorem 28.3 adds an evaluated variance and quantitative, locally resolved statements through the positive representation.

28.4 Cumulants at arbitrary cutoffs

For each fixed positive integer j , the same decomposition gives

$$\kappa_{2j}(X(U_N)) = b_N \kappa_{2j}(Z) + O_{w,j}(1), \quad \kappa_{2j-1}(X(U_N)) = O_{w,j}(1). \quad (28.14)$$

One must justify differentiating here; a pointwise pressure limit alone would not suffice. To do so, combine the block decomposition and the exact random-sum law to write, conditional on the selected block and depth, $X(U_N) = A_N + \sum_{i=1}^{J_N} Y_i$. Both A_N and $J_N - b_N/2$ have uniformly bounded exponential moments in a neighborhood of zero. This follows from the geometric depth tails and the $O(j+1)$ offset bound, including the exponentially rare base component. Therefore

$$\mathbb{E} e^{sX(U_N)} = e^{b_N \chi(s)} \mathbb{E} \exp \{sA_N + (J_N - b_N/2) \log \mathbb{E} e^{sY_1}\}.$$

The last factor is analytic, uniformly bounded, and uniformly nonzero on a sufficiently small disk about 0; it is 1 at 0, and uniform analyticity bounds its derivative there and nearby. All fixed derivatives of its logarithm are bounded. Differentiating proves (28.14). For the run family, with Bernoulli numbers defined by $s/(e^s - 1) = \sum B_j s^j / j!$,

$$\kappa_{2j}(X(U_N)) = b_N \frac{B_{2j}(a^{2j} - 1)}{2j} + O_{a,j}(1). \quad (28.15)$$

29 Exact pressure and the rarity of logarithmic errors

Theorem 29.1 (Pressure for every cutoff). *For every real s ,*

$$\log \mathbb{E} e^{sX(U_N)} = (\log_a N) \chi(s) + O_{w,s}(1), \quad \chi(s) = \frac{1}{2} \log \frac{S(e^s)S(e^{-s})}{a^2}. \quad (29.1)$$

The function χ is even, analytic on the real line, strictly convex, and satisfies $\chi(0) = \chi'(0) = 0$, $\chi''(0) = v$.

Proof. For real s , Cauchy–Schwarz gives $\Delta(e^s) \geq a^2 > 1$. The positive finite formula, and its odd-depth counterpart, give

$$\log F_{r,1}(e^s) = \frac{r}{2} \log \Delta(e^s) + O_{w,s}(1).$$

Let $L_{h-1} < N \leq L_h$. Since W_{h-1} is a prefix of W_h , positivity gives the decisive squeeze

$$F_{h-1,1}(e^s) \leq \sum_{n=0}^{N-1} e^{sX(n)} \leq F_{h,1}(e^s).$$

Subtract $\log N$ and use $h = \log_a N + O_w(1)$. Strict convexity follows because $\chi''(s)$ is one half the sum of the variances of the nondegenerate finite distribution Z under tilts s and $-s$. $\square$

Define the rate function

$$I_w(x) = \sup_{s \in \mathbb{R}} \{sx - \chi(s)\}. \quad (29.2)$$

Let $z_- = \min \text{supp } Z$, $z_+ = \max \text{supp } Z$, $d = z_+ - z_-$, and let c_- , c_+ be the multiplicities of z_- , z_+ among the a zero heights.

Theorem 29.2 (A full large-deviation law). *The variables $X(U_N)/\log_a N$ satisfy a large-deviation principle with speed $\log_a N$ and rate (29.2). Thus for a closed set $F \subseteq \mathbb{R}$ and an open set $G \subseteq \mathbb{R}$,*

$$\limsup_{N \rightarrow \infty} \frac{1}{\log_a N} \log \mathbb{P} \left(\frac{X(U_N)}{\log_a N} \in F \right) \leq - \inf_F I_w, \quad (29.3)$$

$$\liminf_{N \rightarrow \infty} \frac{1}{\log_a N} \log \mathbb{P} \left(\frac{X(U_N)}{\log_a N} \in G \right) \geq - \inf_G I_w. \quad (29.4)$$

Its effective domain is $[-d/2, d/2]$ and its endpoint values are

$$I_w(d/2) = I_w(-d/2) = \log a - \frac{1}{2} \log(c_- c_+). \quad (29.5)$$

For $|x| < d/2$, the supremum is attained at the unique solution $\chi'(s) = x$.

Proof. Exponential Markov bounds from (29.1) give the upper bound on an interval; finitely many such intervals cover a compact closed set. Exponential tightness also follows either from the same bounds or from the deterministic logarithmic bound proved in Section 32.

For the lower bound at $x = \chi'(s)$, tilt the distribution by $e^{sX}/\mathbb{E}e^{sX}$. Under the tilted measure the scaled logarithmic moment generating function is $\chi(s+t) - \chi(s)$. Strict convexity and exponential Markov bounds with small positive and negative t show concentration of $X/\log_a N$ within any fixed neighborhood of x . Undoing the tilt on this neighborhood gives the lower exponent $\chi(s) - sx - |s|\varepsilon$; let $\varepsilon \downarrow 0$. This proves the lower bound for all interior points, and hence for all open sets meeting the interior.

As $s \rightarrow +\infty$,

$$\chi(s) = \frac{d}{2}s - \log a + \frac{1}{2} \log(c_- c_+) + o(1).$$

The analogous formula holds at $-\infty$. It determines the effective domain and the endpoint values. The rate approaches these finite endpoint values from the interior. An open neighborhood of an endpoint contains interior points, completing the lower bound there. $\square$

Remark 29.3 (Endpoint events). The endpoint assertion concerns neighborhoods in a large-deviation principle. It does not assert a single asymptotic formula for one exact extremal integer at every cutoff. Moreover the endpoint rate is $\log a$ only when both extreme zero heights have multiplicity one.

29.1 Explicit formulas in the run family

For Z uniform on $-1, 0, \dots, a-2$, the translation cancels and

$$\chi_a(s) = \log \left(\frac{\sinh(as/2)}{a \sinh(s/2)} \right). \quad (29.6)$$

For $0 < |x| < (a-1)/2$, the tilting parameter is determined by

$$x = \frac{a}{2} \coth \frac{as}{2} - \frac{1}{2} \coth \frac{s}{2}, \quad I_a(x) = sx - \chi_a(s). \quad (29.7)$$

The continuation at $s = 0$ has $x = 0$. The rate is even and

$$I_a(0) = 0, \quad I_a\left(\pm \frac{a-1}{2}\right) = \log a.$$

Near zero,

$$I_a(x) = \frac{6x^2}{a^2 - 1} + \frac{36(a^2 + 1)x^4}{5(a^2 - 1)^3} + O_a(x^6). \quad (29.8)$$

This follows by reverting $\chi'_a(s) = vs + \kappa_4(Z)s^3/6 + O(s^5)$.

For every fixed $0 < x < (a-1)/2$, the large-deviation law yields the exact logarithmic exponent

$$\frac{1}{N} \#\{0 \leq n < N : X_a(n) \geq x \log_a N\} = N^{-I_a(x)/\log a + o(1)}. \quad (29.9)$$

To see the matching lower bound, use a small interval strictly above x and then let its left endpoint decrease to x . The upper bound follows from the closed half-line and the strict increase of I_a on its positive half-domain.

Equation (29.9) describes how exceptional prefixes interpolate between the Gaussian scale and the extremal logarithmic scale. It gives a frequency exponent, not a leading multiplicative constant.

30 Which arrangement has the largest fluctuations?

The incidence matrix (25.3) is the same for all choices of w at fixed a . It determines the frequencies and the substitution growth, but not the magnitude of the fluctuations. The zero-height representation makes the comparison exact.

Theorem 30.1 (Sharp variance and pressure comparison). *For every admissible w ,*

$$\boxed{\frac{a-1}{a^2} \leq v \leq \frac{a^2-1}{12}}. \quad (30.1)$$

The upper equality holds exactly for

$$w = 1^h 0^a 1^{a-1-h}, \quad 1 \leq h \leq a-1. \quad (30.2)$$

The lower equality holds exactly for

$$w = (10)^{a-1} 0 \quad \text{or} \quad w = 10(01)^{a-2} 0. \quad (30.3)$$

They coincide when $a = 2$.

For every real s the entire pressure is bounded by

$$\frac{1}{2} \log \left(1 + \frac{2(a-1)}{a^2} (\cosh s - 1) \right) \leq \chi_w(s) \leq \log \left(\frac{\sinh(as/2)}{a \sinh(s/2)} \right). \quad (30.4)$$

At any $s \neq 0$, its equality cases are exactly the corresponding words in (30.3) and (30.2).

Proof. Sort the a zero heights, with multiplicity, as $z_1 \leq \dots \leq z_a$. By Lemma 25.2, each successive gap is either 0 or 1. Hence $0 \leq z_j - z_i \leq j - i$, and

$$v = \frac{1}{a^2} \sum_{i < j} (z_j - z_i)^2 \leq \frac{1}{a^2} \sum_{i < j} (j - i)^2 = \frac{a^2 - 1}{12}.$$

Equality requires all successive gaps to be 1, so all heights are distinct. An internal run of ones between two zeros forces a repetition of a zero height: its downward move either lands inside the already crossed interval, or lands below that interval and must later cross it again to end at height 1. Therefore distinct heights force all zeros into one run. Conversely, (30.2) gives a consecutive distinct heights and attains the bound.

A nonconstant list of a integers has at least $a - 1$ unequal unordered pairs, each contributing at least 1 to the variance sum. This gives the lower bound. Equality requires two adjacent heights with multiplicities 1 and $a - 1$. Since -1 and 0 are both present, these are exactly the two heights. Such a path consists of excursions 10 below height 0 and 01 above it, followed by its last 0. It starts with a 10 excursion. Requiring one of the two heights to occur only once gives exactly (30.3).

Finally,

$$\Delta(e^s) = a^2 + 2 \sum_{i < j} (\cosh(s(z_j - z_i)) - 1). \quad (30.5)$$

The same pairwise upper and lower comparisons give

$$a^2 + 2(a - 1)(\cosh s - 1) \leq \Delta(e^s) \leq \left(\frac{\sinh(as/2)}{\sinh(s/2)} \right)^2.$$

Take one half the logarithm after division by a^2 . For $s \neq 0$, strict increase of $\cosh(sx)$ in $x \geq 0$ gives precisely the already identified equality cases. $\square$

Corollary 30.2 (Ordering the deviation costs). *Let I_{run} be the rate function for any word in (30.2), and I_{alt} that for either word in (30.3). Then, with extended-real values allowed,*

$$I_{\text{run}}(x) \leq I_w(x) \leq I_{\text{alt}}(x) \quad (x \in \mathbb{R}).$$

Proof. Taking a Legendre transform reverses pointwise inequalities. $\square$

Thus a single zero run maximizes more than the variance: it maximizes the complete logarithmic moment generating function. It minimizes the exponential cost of every deviation on the common part of the effective domains. The critical family selected by the repository is an extremal member of a much larger class.

31 Which height profiles occur? A second OEIS connection

The distribution of the zero heights is sufficient to recover the complete histograms: it determines S , then $T = t(S - 1)$, the seeds, and every expression in Theorem 26.2. This makes the inverse problem natural. Are there constraints beyond positivity, total mass, and the no-gap condition?

Theorem 31.1 (Complete realization and profile count). *Fix $a \geq 2$. A Laurent polynomial is the zero-height polynomial of an admissible word if and only if it has the form*

$$S(t) = \sum_{h=\ell}^H c_h t^h, \quad c_h \in \{1, 2, \dots\}, \quad \sum_{h=\ell}^H c_h = a, \quad \ell \leq -1 < 0 \leq H.$$

The number of distinct such polynomials is

$$\sum_{k=2}^a (k-1) \binom{a-1}{k-1} = (a-1)2^{a-2}. \quad (31.1)$$

Thus these profiles give an interpretation of A001787($a - 1$), where A001787 is $n2^{n-1}$ with its usual value 0 at $n = 0$ [12].

Proof. Necessity is Lemma 25.2. For sufficiency, construct a directed multigraph on the integers $\ell, \dots, H + 1$. For each h put c_h upward edges $h \rightarrow h + 1$ and $c_h - \mathbf{1}_{\{h=0\}}$ downward edges $h + 1 \rightarrow h$. There are a upward and $a - 1$ downward edges. The difference between outdegree and indegree is 1 at 0, -1 at 1, and zero at every other vertex.

There is a downward edge $0 \rightarrow -1$, because $c_{-1} > 0$. Prescribe this as the first edge and temporarily remove it. All upward edges remain, so the underlying undirected graph is still connected. The new degree imbalance is 1 at -1 and -1 at 1. The directed Euler-trail criterion therefore gives a trail from -1 to 1 using all remaining edges. Prepending the removed edge gives a trail from 0 to 1 beginning with a downward step. Read upward edges as 0 and downward edges as 1. The resulting word begins with 1, has the required letter counts, and has exactly c_h zero occurrences beginning at height h .

If the support has k values, there are $k - 1$ positions for a consecutive integer interval containing both -1 and 0. Its positive coefficient vectors are the $\binom{a-1}{k-1}$ compositions of a into k parts. Summing over k proves the first equality in (31.1). The second is the standard binomial identity $\sum_{j=0}^{a-1} j \binom{a-1}{j} = (a-1)2^{a-2}$. $\square$

This counts profiles, not words. There are $\binom{2a-2}{a}$ admissible words, and different words can have the same profile. Such words have identical complete-block prefix histograms at every depth, even when their internal prefix order differs. Arbitrary-cutoff bounded corrections may retain information that the profile forgets. For $a = 2, \dots, 9$ the profile counts are

$$1, 4, 12, 32, 80, 192, 448, 1024.$$

The constructive proof also supplies an algorithm for generating a representative word for every profile.

32 Sharp logarithmic envelopes and exact first hits

32.1 A sharp constant for every arrangement

Theorem 32.1 (General critical cluster interval). *For an arbitrary admissible w , let d be the diameter of its zero-height support. Then*

$$\limsup_{N \rightarrow \infty} \frac{X(N)}{\log N} = -\liminf_{N \rightarrow \infty} \frac{X(N)}{\log N} = \frac{d}{2 \log a}. \quad (32.1)$$

The complete cluster set is the closed interval between these endpoints. In particular $|X(N)| \leq (d/2) \log_a N + O_w(1)$.

Proof. The support of Δ is the full integer interval $[-d, d]$, and both endpoint coefficients are positive. Since each K_c is a fixed nonzero positive polynomial, (26.6) shows that the maximum and minimum proper-prefix weights at depth r are $dr/2 + O_w(1)$ and $-dr/2 + O_w(1)$. The odd-depth identity supplies the same statement at odd r . Enclosing an arbitrary prefix between adjacent full-word lengths proves the uniform bound.

Choose positions p_r, q_r attaining the maximum and minimum in a depth- r block. The uniform bound applied to their attained values forces $\log_a p_r = r + O_w(1)$ and $\log_a q_r = r + O_w(1)$. The opposite inequalities follow from $p_r, q_r \leq L_r$. These positions attain both constants in (32.1). Between p_r and q_r the walk changes by ± 1 at every step and therefore crosses every intermediate integer height. All these positions have logarithm $r \log a + O_w(1)$. Selecting crossings of any prescribed proportional height gives every intermediate normalized limit. $\square$

32.2 Exact finite extrema in the repository's critical family

Now specialize to (24.3), so $a \geq 3$ and $d = a - 1$. Set $W_{-1} = 0$. Its word recursion is

$$W_{r+1} = W_r W_{r-1}^a W_r^{a-2}, \quad X(W_r) = (-1)^{r+1}. \quad (32.2)$$

Proposition 32.2 (Finite envelopes). *Including both empty and complete prefixes of W_r , the maximum and minimum weights are*

$$A_r = \left\lceil \frac{r}{2} \right\rceil (a - 1), \quad B_r = - \left\lfloor \frac{r}{2} \right\rfloor (a - 1) - 1 \quad (r \geq 0). \quad (32.3)$$

Every integer in $[B_r, A_r]$ occurs.

Proof. The initial values are $A_0 = 0$, $B_0 = -1$, $A_1 = a - 1$, and $B_1 = -1$. The offsets of the blocks in (32.2) give, for odd $r \geq 1$,

$$A_{r+1} = \max(A_r, A_{r-1} + 1), \quad B_{r+1} = \min(B_r + 1 - a, B_{r-1} + 2 - a),$$

and for even $r \geq 1$,

$$A_{r+1} = \max(A_r + a - 1, A_{r-1} + a - 2), \quad B_{r+1} = \min(B_r, B_{r-1} - 1).$$

Substitution proves (32.3) by induction. The nearest-neighbor property proves the interval assertion. Alternatively, at positive even depths the outer exponents of the histogram give the same extrema directly. $\square$

Theorem 32.3 (Explicit first extremal prefixes). *Define*

$$P_0 = \emptyset, \quad P_{k+1} = \sigma_a^2(P_k) 1 0^a, \quad Q_k = \sigma_a(P_k) 1. \quad (32.4)$$

Then P_k and Q_k are the first prefixes of the infinite word with weights $k(a - 1)$ and $-k(a - 1) - 1$, respectively. Writing

$$A = \frac{4a}{(a - 1)(a + 1)^2}, \quad B = \frac{(a - 1)^2}{a + 1}, \quad (32.5)$$

their exact lengths are

$$p_k = A(a^{2k} - 1) + Bk, \quad q_k = aA(a^{2k} - 1) - Bk + 1. \quad (32.6)$$

Proof. Every P_k is a legal prefix immediately before a one; when $k \geq 1$ it ends in zero. The word $1 0^a$ is a prefix immediately before a one in $\sigma_a^2(1)$, so induction proves legality. Every Q_k ends in a one immediately before a zero. The eigenweight identity gives the stated weights.

For $k \geq 1$, P_k is a proper prefix of W_{2k-1} , and Q_k is a proper prefix of W_{2k} . The extreme coefficients of Δ and K_1 are all 1 in this family. Equation (26.6) therefore shows that each extreme of W_{2k} is attained at exactly one proper prefix. The full word has weight -1 and adds no duplicate extreme. The exhibited prefixes consequently are the first occurrences of their weights in the infinite fixed point.

For a finite word P ,

$$|\sigma_a(P)| = a|P| - (a - 1)X(P), \quad |\sigma_a^2(P)| = a^2|P| - (a - 1)^2X(P).$$

Thus

$$p_{k+1} = a^2 p_k - (a - 1)^3 k + a + 1, \quad q_k = a p_k - (a - 1)^2 k + 1.$$

Solve the first recurrence with $p_0 = 0$ and substitute into the second to obtain (32.6). $\square$

In particular $p_1 = a + 1$, $q_1 = 3a$, $p_k \sim Aa^{2k}$ and $q_k \sim aAa^{2k}$. At $a = 3$ the first positive hitting lengths are 4, 32, 276, ... and the negative ones are 9, 89, 817, ... These formulas answer an explicit extremal slice of the predecessor's first-hitting question as well as its critical normalization question.

32.3 Counting and position errors

Let $O(N)$ count ones and $Z_0(N)$ count zeros. Then

$$Z_0(N) - N/2 = X(N)/2, \quad O(N) - N/2 = -X(N)/2.$$

Both frequencies are $1/2$. For the run family their normalized counting-error cluster intervals are $[-(a-1)/(4 \log a), (a-1)/(4 \log a)]$.

If $u(n)$ and $v(n)$ are the positions of the n th zero and one, respectively, their errors satisfy

$$E_0(n) = 2n - u(n) = X(u(n)), \quad E_1(n) = 2n - v(n) = -X(v(n)). \quad (32.7)$$

Thus $u(n) = 2n + O_a(\log n)$ and $v(n) = 2n + O_a(\log n)$. Both $E_i(n)/\log n$ have the complete cluster interval

$$\left[-\frac{a-1}{2 \log a}, \frac{a-1}{2 \log a} \right]. \quad (32.8)$$

To justify transfer of every intermediate value, both letters occur within a bounded distance of every position: each of the two fixed finite words $\sigma_a^2(0), \sigma_a^2(1)$ contains both letters, and the fixed point is their concatenation. Move a chosen prefix endpoint to the next required letter. The length and weight change by $O_a(1)$, which does not change the logarithmic limit.

The four explicit extremal rank sequences, with $h_k = k(a-1)$, are as follows.

Prefix	Last letter	Rank	Position error
P_k	0	$(p_k + h_k)/2$	$E_0 = h_k$
$Q_k 0$	0	$(q_k + 1 - h_k)/2$	$E_0 = -h_k$
Q_k	1	$(q_k + h_k + 1)/2$	$E_1 = h_k + 1$
$P_k 1$	1	$(p_k - h_k + 2)/2$	$E_1 = 1 - h_k$

32.4 An exact letter-conditioned refinement

The paired matrix $C(t)^m$ also records the histogram separately according to the next letter. Its (c, j) entry counts the proper prefixes of $\sigma^{2m}(c)$ immediately followed by j . Moreover

$$C - I = \begin{pmatrix} S-1 \\ T-S \end{pmatrix} \begin{pmatrix} 1 & t \end{pmatrix}. \quad (32.9)$$

All its entries are nonnegative: S contains the zero-height upward crossing, so $S-1 \geq 0$. Consequently

$$C^m = I + (C - I) \sum_{h=0}^{m-1} \Delta^h \quad (32.10)$$

is a positive histogram formula even after conditioning on the next letter. The two seed laws in each row differ by a shift of one. This supplies exact distributions for the position errors at complete substitution cutoffs, through $E_0 = X(\text{preceding prefix}) + 1$ and $E_1 = 1 - X(\text{preceding prefix})$. Their bulk variance and pressure are the same as above.

33 An exact supercritical amplitude and its self-affine profile

In this section we return to the two-parameter family of the predecessor. The weight is now $\#0 - q\#1$, and $L(P)$ denotes geometric length; ordinary word length remains $|P|$.

Let $a, b \geq 1$ be integers with $a \geq b + 3$, and let w be the fixed point starting with 1 of

$$\tau(0) = 1, \quad \tau(1) = 10^a 1^b.$$

Write

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2} > 1, \quad \lambda = q + b + 1 = \frac{a}{q}, \quad \alpha = \frac{\log q}{\log \lambda} \in (0, 1).$$

For a finite word P , define

$$X(P) = |P|_0 - q|P|_1, \quad L(P) = |P|_0 + \lambda|P|_1.$$

Both are additive, and

$$X(\tau(P)) = -qX(P), \quad L(\tau(P)) = \lambda L(P).$$

For the first N letters of w , abbreviate their weight and geometric length by $X(N)$ and L_N .

Theorem 33.1 (Exact supercritical normalization). *Set*

$$C_{a,b} = \frac{\lambda - 1}{q^2 - 1} \left(\frac{(\lambda + q)(\lambda^2 - 1)}{(q + 1)^2} \right)^\alpha.$$

Then

$$\limsup_{N \rightarrow \infty} \frac{X(N)}{N^\alpha} = C_{a,b}, \quad \liminf_{N \rightarrow \infty} \frac{X(N)}{N^\alpha} = -C_{a,b}.$$

The full cluster set is $[-C_{a,b}, C_{a,b}]$. There is a continuous function $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $\Theta(s + 1) = -\Theta(s)$ such that, with $c = (\lambda + q)/(q + 1)$,

$$X(N) = c^\alpha N^\alpha \Theta(\log_\lambda(cN)) + O_{a,b}(N^{\alpha^2}).$$

The function is obtained from a two-state graph-directed system with explicit affine maps. Its amplitude is

$$\max_s |\Theta(s)| = \frac{\lambda - 1}{q^2 - 1} \left(\frac{\lambda^2 - 1}{q + 1} \right)^\alpha.$$

33.1 Continuous block profiles

Assign geometric letter lengths $\ell_0 = 1, \ell_1 = \lambda$ and vertical increments $\chi_0 = 1, \chi_1 = -q$. Let $H_{i,k}$ be the polygonal path of $\tau^k(i)$ with these increments. Define

$$f_{i,k}(t) = (-q)^{-k} H_{i,k}(\lambda^k t), \quad 0 \leq t \leq \ell_i.$$

The initial profiles are $f_{0,0}(t) = t$ and $f_{1,0}(t) = -qt/\lambda$. The pair of continuous functions with endpoint data $f_i(0) = 0, f_i(\ell_i) = \chi_i$ is a complete metric space in the maximum uniform norm. On the interval corresponding to a letter j of $\tau(i)$, preceded by a prefix P of that image, set

$$(Tf)_i \left(\frac{L(P) + u}{\lambda} \right) = -\frac{X(P) + f_j(u)}{q}, \quad 0 \leq u \leq \ell_j.$$

Endpoint compatibility makes this a continuous function. The operator T is a contraction of ratio $1/q$, and $f_{\cdot,k+1} = Tf_{\cdot,k}$. Consequently the profiles converge uniformly to a unique pair (f_0, f_1) . Its graph maps are exactly

$$(u, y) \mapsto \left(\frac{L(P) + u}{\lambda}, -\frac{X(P) + y}{q} \right).$$

In particular they contract in both coordinates.

For later verification, their branch equations are

$$\begin{aligned} f_0(t) &= -q^{-1} f_1(\lambda t), & 0 \leq t \leq 1, \\ f_1(t) &= -q^{-1} f_1(\lambda t), & 0 \leq t \leq 1, \\ f_1\left(1 + \frac{j+u}{\lambda}\right) &= 1 - \frac{j}{q} - \frac{f_0(u)}{q}, & 0 \leq j < a, \quad 0 \leq u \leq 1, \\ f_1\left(q + 1 + \ell + \frac{u}{\lambda}\right) &= 1 - \lambda + \ell - \frac{f_1(u)}{q}, & 0 \leq \ell < b, \quad 0 \leq u \leq \lambda. \end{aligned}$$

The first-level difference from the initial line is linear on each piece. For state 0 it vanishes. For state 1 its largest absolute value occurs after the zero block and equals

$$D_0 = \frac{b(\lambda + q)}{\lambda}.$$

Therefore, with

$$E = \frac{bq(\lambda + q)}{\lambda(q - 1)},$$

the uniform error satisfies

$$\max_i \|f_{i,k} - f_i\|_\infty \leq Eq^{-k}.$$

Let $H : [0, \infty) \rightarrow \mathbb{R}$ be the corresponding polygonal path of the infinite fixed point. Because $\tau^k(1)$ is a prefix of w , $f_{1,k}(t) = (-q)^{-k} H(\lambda^k t)$ for $0 \leq t \leq \lambda$. Extend the limiting profile by

$$F(t) = (-q)^k f_1(t/\lambda^k)$$

for any sufficiently large integer $k \geq 0$. The first branch equation makes this independent of k . It gives a continuous function satisfying

$$F(\lambda t) = -qF(t), \quad |H(t) - F(t)| \leq E \quad (t \geq 0).$$

Every refinement preserves all existing polygon vertices, since the child profiles have fixed endpoint weights. Thus $f_1 = f_{1,k}$ at the depth- k vertices. Choosing a substitution prefix long enough to contain a given integer prefix gives the exact interpolation identity

$$F(L_N) = X(N) \quad (N \geq 0).$$

The bounded-error comparison with H is only needed between these vertices.

The function F is globally Hölder continuous of exponent α . Indeed, at depth k the profile intervals have lengths $\ell_i \lambda^{-k} \geq \lambda^{-k}$ and oscillations bounded by Dq^{-k} , where $D = \max_i \text{osc}(f_i)$. An interval of length at most λ^{-k} meets at most two interiors of depth- k cells; allowing three cells avoids all endpoint conventions. Choosing k at the scale of $|x - y|$ proves a uniform α -Hölder bound on $[0, \lambda]$. The relation $F(\lambda t) = -qF(t)$ extends the same bound to all positive arguments, since $q = \lambda^\alpha$.

33.2 A concave polygon envelope: proof of the exact amplitude

This is the substantive sharp-constant argument. Put

$$d = q + 1, \quad e = \lambda - 1, \quad r = \lambda^{-2}, \quad s = q^{-2}, \quad t_* = \frac{d}{1-r}, \quad A = \frac{e}{1-s}.$$

Thus $0 < r < s < 1$, $s = r^\alpha$, and

$$d = (1 - r)t_*, \quad e = (1 - s)A.$$

Moreover,

$$\frac{t_*}{\lambda} < 1 < t_*, \quad rt_* < 1,$$

because $\lambda - q - 1 = b \geq 1$ and $\lambda^2 - 1 - \lambda(q + 1) = b\lambda - 1 > 0$.

Define $P : [0, \infty) \rightarrow [0, A]$ by

$$P(0) = 0, \quad P(t_* r^n) = As^n \quad (n \geq 0),$$

using linear interpolation between consecutive displayed points and setting $P(t) = A$ for $t \geq t_*$. Its successive slopes are

$$m(s/r)^n, \quad m = \frac{A(1-s)}{t_*(1-r)} = \frac{e}{d}.$$

Thus P is increasing and concave, with slope at most m on $[1, \infty)$. Because it interpolates the concave power curve at its breakpoints,

$$P(t) \leq \frac{A}{t_*^\alpha} t^\alpha \quad (t \geq 0).$$

It also satisfies

$$P(d + rv) \geq e + sP(v) \quad (v \geq 0). \quad (33.1)$$

For $v \geq t_*$ both sides equal A . For $v < t_*$, the argument $d + rv$ lies in the last linear segment $[rt_*, t_*]$, because $d > rt_*$ follows from $\lambda^2 > 2$. Hence

$$P(d + rv) = A - mr(t_* - v).$$

Concavity gives $A - P(v) \geq m(t_* - v)$; multiplying by $s > r$ proves (33.1). We shall also use

$$q^{-2}P(\lambda^2 t) \leq P(t), \quad \frac{m}{\lambda} < \frac{1}{q}.$$

The first follows from the exact small-scale relation below the final breakpoint and monotonicity above it. The second is $q(\lambda - 1) < \lambda(q + 1)$.

We claim that T preserves the following closed bounds:

$$\begin{aligned} -P(t) &\leq f_1(t) \leq q^{-1}P(\lambda t), \\ -q^{-2}P(\lambda^2 t) &\leq f_0(t) \leq q^{-1}P(\lambda t). \end{aligned}$$

The state-0 equation proves its bounds directly. The first state-1 branch then follows using $q^{-2}P(\lambda^2 t) \leq P(t)$.

For a zero-block branch let $v = \lambda u$ and $t = 1 + j/\lambda + rv$. The lower bound is equivalent to

$$P(t) \geq j/q - 1 + sP(v).$$

Since $t \geq 1$ and $d + rv - t = (a - j)/\lambda$, the slope bound and (33.1) give

$$\begin{aligned} P(t) &\geq P(d + rv) - m(a - j)/\lambda \\ &\geq e + sP(v) - (a - j)/q \\ &= j/q - 1 + sP(v). \end{aligned}$$

For the upper bound, $t \geq 1 > t_*/\lambda$, so its desired value is the constant A/q . The old lower bound on f_0 gives

$$1 - j/q - f_0(u)/q \leq 1 + A/q^3 \leq A/q,$$

where the last inequality is equivalent to $e/q \geq 1$, true since $e = q + b$.

For a trailing-one branch write $v = \lambda u$ and $t = d + \ell + rv$. Its lower bound follows from

$$P(t) \geq P(d + rv) \geq e + sP(v) \geq e - \ell + sP(v).$$

Its upper bound follows from $-e + \ell + P(u)/q \leq A/q$, since $\ell \leq b - 1 < e$. This verifies every branch.

The initial straight profiles belong to these bounds: P is concave, $P(\lambda) = A$, and $A/q > 1$, so $P(\lambda t) \geq At \geq qt$ for $0 \leq t \leq 1$; the state-1 lower bound follows in the same way on $[0, \lambda]$. Contraction therefore gives the asserted bounds for the limit profiles. In particular,

$$|f_1(t)| \leq Kt^\alpha, \quad K = \frac{A}{t_*^\alpha}.$$

In fact the same bound holds for every finite profile $f_{1,k}$, since the bounds contain the initial profiles and are invariant under T . Rescaling a sufficiently long substitution prefix therefore gives the useful exact finite inequality

$$|H(t)| \leq Kt^\alpha \quad (t \geq 0), \quad |X(N)| \leq KL_N^\alpha.$$

This is an inequality in geometric length; the corresponding symmetric physical-length constant need not hold without a lower-order correction at every finite N .

The constant is attained. Set $s_* = t_*/\lambda < 1$. The first and first trailing branch equations give

$$f_1(s_*) = -q^{-1}f_1(t_*), \quad f_1(t_*) = -e - q^{-1}f_1(s_*).$$

Here $t_* = d + t_*/\lambda^2$ ensures that the second evaluation is legal. Solving gives $f_1(t_*) = -A$ and $f_1(s_*) = A/q$. Consequently

$$\sup_{t>0} \frac{|F(t)|}{t^\alpha} = K = \frac{\lambda - 1}{q^2 - 1} \left(\frac{\lambda^2 - 1}{q + 1} \right)^\alpha.$$

Both signs attain this normalized amplitude at geometrically rescaled copies of t_* and s_* .

33.3 Conversion from geometric time to prefix length

Solving for the letter counts gives the exact identities

$$N = \frac{q+1}{\lambda+q}L_N + \frac{\lambda-1}{\lambda+q}X(N), \quad L_N = cN - \frac{\lambda-1}{q+1}X(N).$$

The identity $X(N) = F(L_N)$ and the sharp bound on F first give $X(N) = O(N^\alpha)$, because $N \leq L_N \leq \lambda N$. The Hölder bound then yields

$$X(N) = F(cN) + O(N^{\alpha^2}).$$

Define

$$\Theta(s) = \lambda^{-\alpha s} F(\lambda^s).$$

This function is continuous and anti-periodic with step 1. Its range over a full period is $[-K, K]$: it is a continuous interval, the sharp bound places it inside that interval, and the two explicitly exhibited points attain both endpoints. Every phase is approached by integer subsequences N_k nearest to $c^{-1}\lambda^{2k+s}$. Therefore the full cluster set of $X(N)/N^\alpha$ is $[-c^\alpha K, c^\alpha K]$. The identity $c^\alpha K = C_{a,b}$ proves the theorem.

33.4 Explicit extremizing prefixes and a second check of constants

Define legal prefixes immediately before a one by

$$P_0 = \emptyset, \quad P_{k+1} = \tau^2(P_k) 1 0^a.$$

Their exact weights and geometric lengths are

$$X(P_k) = \frac{q(\lambda - 1)}{q^2 - 1}(q^{2k} - 1), \quad L(P_k) = \frac{\lambda(q + 1)}{\lambda^2 - 1}(\lambda^{2k} - 1).$$

Their physical lengths consequently are

$$\begin{aligned} |P_k| &= \frac{\lambda(q + 1)^2}{(\lambda + q)(\lambda^2 - 1)}(\lambda^{2k} - 1) \\ &\quad + \frac{q(\lambda - 1)^2}{(\lambda + q)(q^2 - 1)}(q^{2k} - 1). \end{aligned}$$

Since $q < \lambda$ and $\lambda^\alpha = q$, direct division verifies

$$\frac{X(P_k)}{|P_k|^\alpha} \rightarrow C_{a,b}.$$

The legal prefixes $Q_k = \tau(P_k)$ satisfy $X(Q_k) = -qX(P_k)$ and $|Q_k|/|P_k| \rightarrow \lambda$, so their normalized weights tend to $-C_{a,b}$.

Scope and priority. The continuous substitution-profile construction and broad spectral power-law phenomenon have general predecessors. The result established here is the exact amplitude for this specified family, with a branchwise concave-envelope certificate and complete physical-length cluster set. No claim of a previously unknown general limit-profile theorem is required. The graph is a graph-directed self-affine object; its scalar normalized cluster set is an interval and should not be described as a Cantor set.

33.5 A square-root example with an exact finite inequality

Take

$$\tau(0) = 1, \quad \tau(1) = 1 0^8 1.$$

Then $q = 2$, $\lambda = 4$, $\alpha = 1/2$, $c = 2$, and $K = \sqrt{5}$. The sharp asymptotic statement becomes

$$\text{Clust}_{N \rightarrow \infty} \frac{\#0 - 2\#1}{\sqrt{N}} = [-\sqrt{10}, \sqrt{10}]. \quad (33.2)$$

There is also a particularly simple statement valid at every finite prefix. Since $L_N = 2N - X(N)$, the geometric inequality $X(N)^2 \leq 5L_N$ is exactly

$$\boxed{X(N)^2 + 5X(N) \leq 10N.} \quad (33.3)$$

Equivalently, $|X(N) + 5/2| \leq \sqrt{10N + 25/4}$. This explains why a finite negative ratio $X(N)/\sqrt{N}$ can lie below $-\sqrt{10}$ while converging to the proved asymptotic envelope.

The positive extremizing prefixes have

$$X(P_k) = 2(4^k - 1), \quad |P_k| = \frac{2 \cdot 16^k + 5 \cdot 4^k - 7}{5}.$$

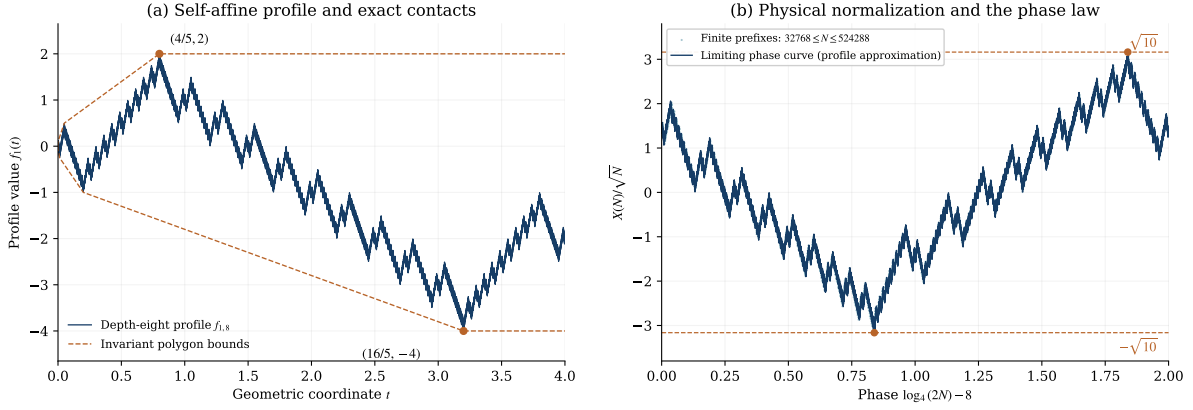


Figure 4: The supercritical example $(a, b) = (8, 1)$. Left: the depth-eight geometric profile, whose uniform distance from f_1 is at most $3/256$, with the invariant polygon bounds and exact limiting contacts $(4/5, 2)$ and $(16/5, -4)$. Right: finite prefix ratios and the limiting phase curve. The dashed levels are the asymptotic physical amplitudes $\pm\sqrt{10}$; the exact finite envelope is (33.3).

The first lengths are 9, 117, 1701, 26469, 420453. Their ratios increase toward $\sqrt{10}$; the asymptotic limit follows directly from these formulas, without a numerical fit.

More generally, for an integer $q \geq 2$, the parameters $a = q^3$, $b = q^2 - q - 1$ give $\lambda = q^2$ and $\alpha = 1/2$. Direct substitution in the theorem yields

$$C_{a,b} = \sqrt{q(q-1)(q^2+1)}$$

and the exact finite inequality

$$X(N)^2 + (q-1)^2(q^2+1)X(N) \leq q(q-1)(q^2+1)N. \quad (33.4)$$

Indeed here $K^2 = (q-1)(q^2+1)$, $c = q$, and $L_N = qN - (q-1)X(N)$. This is an infinite family of explicit quadratic envelopes.

33.6 Consequences for phase-dependent empirical distributions

The continuous profile determines more than the cluster interval. For $n \geq 1$ put $Y(n) = X(n)/n^\alpha$ and $G(s) = c^\alpha \Theta(s)$, a continuous function of period 2. Theorem 33.1 gives

$$Y(n) = G(\log_\lambda(cn)) + O_{a,b}(n^{-\alpha(1-\alpha)}). \quad (33.5)$$

Corollary 33.2 (Logarithmic sampling and phase subsequences). *Let $H_M = \sum_{n=1}^M 1/n$. The logarithmically weighted empirical probability measures satisfy*

$$\frac{1}{H_M} \sum_{n=1}^M \frac{1}{n} \delta_{Y(n)} \implies \nu, \quad \int h \, d\nu = \frac{1}{2} \int_0^2 h(G(s)) \, ds \quad (33.6)$$

for every bounded continuous function h . The limit is symmetric and has support $[-C_{a,b}, C_{a,b}]$.

If $M_j \rightarrow \infty$ and $\log_\lambda(cM_j) \bmod 2 \rightarrow s$, then ordinary uniform sampling has the subsequential law

$$\frac{1}{M_j} \sum_{n=1}^{M_j} \delta_{Y(n)} \implies \text{Law}(G(s + \log_\lambda U)), \quad U \sim \text{Uniform}(0, 1). \quad (33.7)$$

These formulas describe all subsequential laws, allowing that different phases might induce the same measure.

Proof. The error in (33.5) tends to zero and all the variables are bounded, so it can be removed against a continuous test function. In the logarithmic average, uniform continuity allows replacement of the sum by

$$\frac{1}{\log M} \int_1^M h(G(\log_\lambda(cu))) \frac{du}{u}.$$

The substitution $s = \log_\lambda(cu)$ turns this into an average of a continuous function of period 2 over an interval whose length tends to infinity. This proves (33.6). Anti-periodicity gives symmetry. Continuity and the attained extreme values show that every open interval meeting the range has a nonempty open preimage and therefore positive measure.

For (33.7), first restrict n/M_j to $[\varepsilon, 1]$. There the phase approximation is uniform and the sum is a Riemann sum for $h(G(s + \log_\lambda u))$. The omitted proportion is at most ε . Let $\varepsilon \downarrow 0$. Finally every sequence of phases has a convergent subsequence modulo 2, and every prescribed phase is approached by integer cutoffs nearest to $c^{-1}\lambda^{2j+s}$. $\square$

No absolute continuity or absence of atoms is asserted for these pushforward measures. Those properties depend on the level sets and regularity of the profile and are natural next questions.

34 Exact computations and how to reproduce them

The accompanying archive contains the complete $\text{\LaTeX}$ source, the compiled article, four figures, exact data, and four verification programs. The computations have two roles: checking algebraic identities against literal words at feasible sizes, and displaying the exact histogram laws far beyond feasible literal expansion. The proofs above do not depend on a guessed pattern from these checks. [Added 6 October 2026, batch 114: in this report the manuscript's $\text{\LaTeX}$ source and compiled article are not shipped (their text is Part III); the four figures, the data and the four programs are shipped byte-identical with the prefix `03-fluct-`, and the program names below are the shipped names. The programs still write their outputs under their delivery names (Section 23 and the README).]

34.1 What was checked

Program	Scope of completed checks
<code>code/03-fluct-verify.py</code>	37,129 passed integer or rational assertions. Canonical kernels and transfer identities for $3 \leq a \leq 20$; even histograms for $3 \leq a \leq 9$, $0 \leq m \leq 32$; 106 literal words totaling 906,168 symbols. All 4,081 admissible words for $2 \leq a \leq 8$ were checked at depth two; 341 were also checked at depth four.
<code>code/03-fluct-general_word_audit.py</code>	An independent implementation checked all 1,078 words for $2 \leq a \leq 7$, with 4,312 literal histogram comparisons and 4,312 exact finite mean/variance comparisons.
<code>code/03-fluct-verify_profile_realization.py</code>	Constructed all 1,793 profiles for $2 \leq a \leq 9$ and independently enumerated all 15,521 admissible words. The two sets of profiles agreed exactly at each a .
<code>code/03-fluct-verify_supercritical.py</code>	Checked all 524,801 prefixes, including the empty prefix, of $\tau^9(1)$ for $(a, b) = (8, 1)$ against the integer inequality $X^2 + 5X \leq 10N$, and verified the first five explicit extremizing prefixes and their length formulas.

These totals describe finite execution scopes. They do not certify unbounded quantifiers, interval enclosures for floating-point diagnostics, or a formalization in a proof assistant. The separate implementations help detect indexing, parity, and normalization mistakes that a symbolic derivation can conceal.

34.2 Exact histograms at very large lengths

For $a = 3$, the histogram of W_{256} is obtained with only 514 possible integer weights, although the word length is about 2.0851×10^{122} . Every histogram coefficient is computed as an integer; no word of that length is constructed.

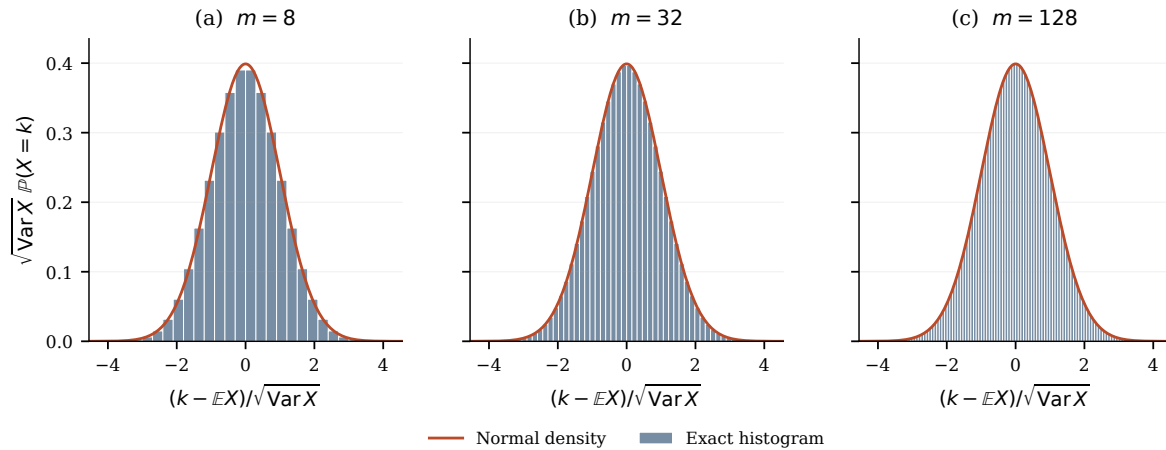


Figure 5: Exact histograms for the critical word $0 \mapsto 1, 1 \mapsto 10001$, at depths $r = 16, 64, 256$. The horizontal coordinate is $(k - \mathbb{E}X_r) / \sqrt{\text{Var } X_r}$; each bar has width $1 / \sqrt{\text{Var } X_r}$ and height $\sqrt{\text{Var } X_r} \mathbb{P}(X_r = k)$. Thus bar area is probability. The orange curve is the standard normal density.

Depth r	Var X_r	CDF error	Scaled local error
16	11.0833334133	0.059691802	0.0042978354
64	43.0833333333	0.030360266	0.0011439270
256	171.083333333	0.015246511	0.0002906199

The CDF error compares both sides of every jump to a normal law using the exact finite mean and variance. The scaled local error is

$$\sup_k \left| \sqrt{\text{Var } X_r} \mathbb{P}(X_r = k) - \varphi\left(\frac{k - \mathbb{E}X_r}{\sqrt{\text{Var } X_r}}\right) \right|.$$

These decimal diagnostics use binary64 arithmetic. They illustrate the proved orders; their printed digits are not interval-certified bounds. The theorem's unscaled full-block local error is $O(r^{-3/2})$, corresponding to a scaled error $O(r^{-1})$.

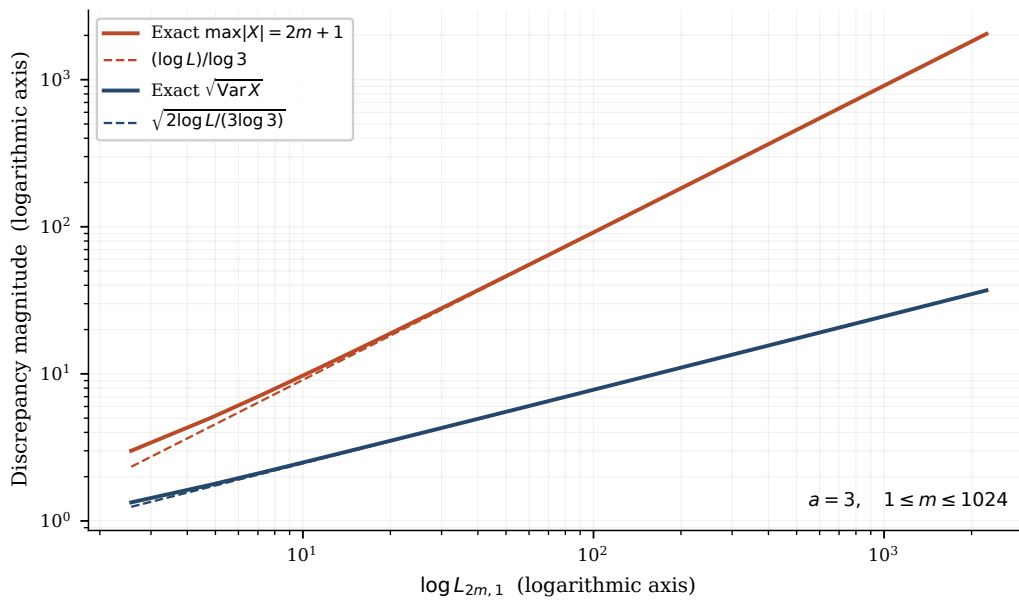


Figure 6: The critical word at $a = 3$ has two different scales: standard deviation proportional to $\sqrt{\log L}$ and largest absolute prefix weight proportional to $\log L$. The horizontal quantity is already $\log L$; both axes are logarithmic. Solid curves use the exact finite expressions and dashed curves their leading terms.

34.3 An economical histogram algorithm

For a fixed word, compute the finite Laurent polynomials S , T , K_0 , K_1 , and Δ . To obtain the even-depth histogram for state c , accumulate

$$H \leftarrow 1, \quad Q \leftarrow K_c; \quad H \leftarrow H + Q, \quad Q \leftarrow Q\Delta \quad \text{for } j = 0, \dots, m-1.$$

The support width grows linearly in m , whereas the literal word length grows like a^{2m} . For a fixed substitution, multiplication by the fixed-support polynomial Δ gives $O_w(m^2)$ integer coefficient operations for this elementary implementation. Coefficient bit lengths are $O_w(m)$; this operation count does not treat arbitrary-size integer arithmetic as having a fixed bit cost. Odd depths use the state-0 identity. The supplied programs implement the formulas independently of literal word generation whenever possible.

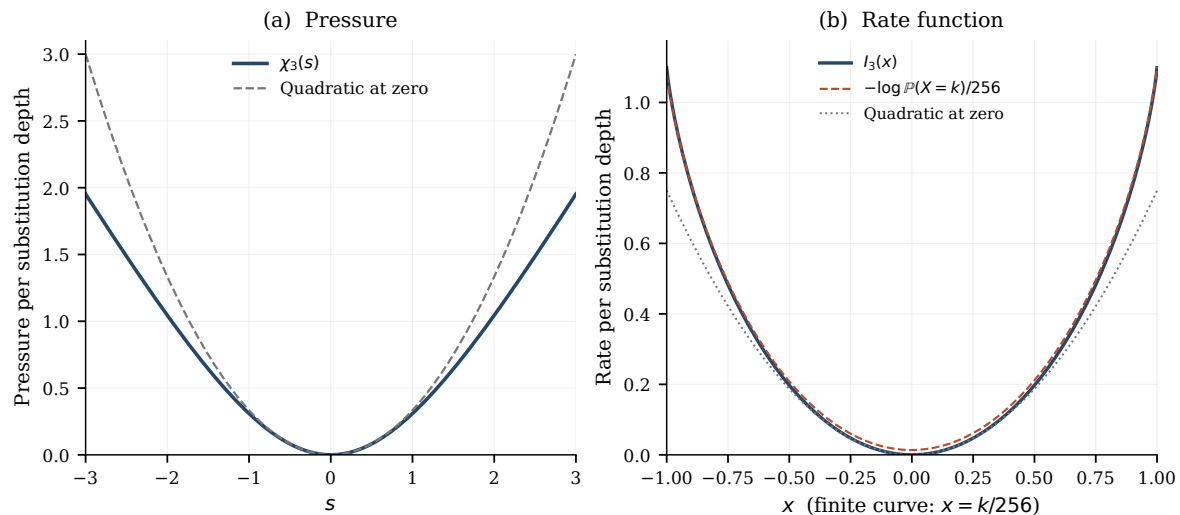


Figure 7: The pressure χ_3 and its Legendre transform I_3 on the substitution-depth normalization used in this article. The finite comparison on the right uses W_{256} , plotting $-\log \mathbb{P}(X_{256} = k)/256$ at $x = k/256$. Its finite offset includes the usual local-mass prefactor. The endpoint rate is $\log 3$ and the limiting domain is $[-1, 1]$.

34.4 Build and verification commands

From the archive directory, the article can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  binary_morphic_fluctuations.tex
```

The bibliography is embedded in the source, so no separate bibliographic database is required. A standard $\text{\TeX}$ installation needs the packages listed in the preamble.

The first three verification commands use only Python’s standard library when figure generation is omitted:

```
python3 code/verify.py --skip-figures
python3 code/general_word_audit.py
python3 code/verify_profile_realization.py
```

To reproduce the figures and the supercritical computation, install the versions recorded in `code/requirements.txt` in an isolated Python environment, then run

```
python3 code/verify.py
python3 code/verify_supercritical.py
```

The JSON files preserve exact integers and rational moments. The text report records the check scopes and the numerical conventions. [Added 6 October 2026, batch 114: these are the commands of the delivered layout. In this report the manuscript source `binary_morphic_fluctuations.tex` is not shipped, so the first command and `code/03-fluct-build.sh` (the delivered `build.sh`) do not apply; the report is built from `article.tex`. The delivered `code/requirements.txt` is shipped as `data/03-fluct-requirements.txt`. The programs resolve their output directories relative to their own location and write `data/` and `figures/` files under unprefix delivery names, so they are to be run on a copy of the delivered layout

(README, “Rerun the checks”). At intake (on such a copy) the three standard-library programs passed in seconds, with the totals of the table above, and `verify_supercritical.py` reported `all_exact_checks_passed: true.`]

35 Further questions and research

The exact representation closes several questions while making others precise. The following projects have distinct mathematical content; none is claimed solved by the present computations. [Added 6 October 2026, batch 114: items 1–10 are the manuscript’s section “Further research: concrete questions after these results”, as delivered. Items 11 and 12 were added at the write under Vladimir’s standing rule of 4 October 2026: they are statements the manuscript makes with an argument sketch only (Section 28). Item 1 overlaps Part II’s research question 2 (Section 21): both ask for the limiting distributions in the bounded regime, item 1 for the prefix weight under uniform prefix sampling and Part II’s for the position errors; item 3 continues Part II’s question 4, of which Theorem 32.3 answers the extremal slice at $a = b + 2$.]

1. **Bounded-regime value distributions.** For the repository’s original cases $a \leq b + 1$, determine the limiting distribution, if any, of $\#0 - q\#1$ under uniform prefix sampling. For $(a, b) = (1, 2)$ and $(2, 1)$, obtain explicit laws for the OEIS words A284368 and A284369 and transfer them to the position sequences. Decide whether the laws have densities, atoms, or a singular continuous part. Our critical positive mixture does not automatically extend across $q = 1$.
2. **Local large deviations with a leading constant.** For the critical class, prove a saddle-point formula for $\mathbb{P}(X_r = k)$ uniformly when k/r lies in a compact subset of $(-d/2, d/2)$, evaluating the prefactor from g_0 and g_1 . Then characterize the additional cutoff dependence when N is not a substitution length. The present pressure gives the exponential cost; the arithmetic phase and exact leading constant remain to be organized.
3. **Extreme atoms and near-extreme multiplicities.** The endpoint rate depends only on c_{-c+} , while the run family has unique even-block extremal prefixes. Classify the exact multiplicities and locations of the first j levels inward from each extreme for a general profile. Determine how much of this information is determined by S , and what further data are needed to recover first-hit positions in the word.
4. **Counting words with a prescribed profile.** Theorem 31.1 counts distinct profiles and constructs one representative. Determine the number of admissible words realizing each coefficient vector (c_h) . The Euler-trail description suggests an exact formula using directed trail enumeration, with care about indistinguishable parallel edges and the prescribed first letter. This would refine the A001787 interpretation and identify the largest histogram equivalence classes.
5. **Inverse problems from fluctuation data.** Determine when two distinct profiles have the same pressure, that is, $S_1(t)S_1(t^{-1}) = S_2(t)S_2(t^{-1})$. Translations and reflection are immediate sources of agreement, but factorization can produce further coincidences. Classify which additional seed or parity data recover S . This separates equality of bulk laws from equality of complete finite histograms.
6. **Regularity and distribution of the supercritical profile.** Determine the sharp local Hölder exponents and the sizes of level sets of Θ . Is the logarithmic-sampling law (33.6) absolutely continuous, atomless, or singular for particular (a, b) ? Obtain a usable density equation where one exists. The scalar cluster set is already known to be an interval; the unresolved geometry concerns the graph and its induced measures.
7. **Sharper physical-time asymptotics.** The conversion from geometric length to letter count gives an error $O(N^{\alpha^2})$ using global Hölder continuity. Find whether that exponent is sharp, and identify parameters or phases where a smaller error holds. A second-order expansion

may require local information about Θ rather than a derivative, because differentiability is not assumed.

8. **Optimizing supercritical arrangements.** Keep the incidence matrix fixed but rearrange the zeros and ones in the long image. Determine which arrangements maximize or minimize the normalized supercritical amplitude. The critical variance comparison uses ordered zero heights; its supercritical analogue involves a discounted branch optimization. Seek an algorithm that outputs a verifiable polygon or other finite envelope certificate.
9. **Higher alphabets and roots of unity.** For primitive substitutions with an eigenvalue that is a root of unity of order greater than two, group that many levels and ask when the histogram transfer matrix admits a nonnegative low-rank remainder. A useful target is an exact finite mixture of independent vector increments. The invariant-section identity by itself does not establish this positivity.
10. **Formal proof and certified computation.** Formalize the positive histogram identity and its finite moment consequences first: they involve finite words, Laurent polynomials, and elementary sums. A separate formalization could check the finitely many branch inequalities of the supercritical envelope. For publication-quality numerical constants beyond the closed forms here, add interval-certified saddle-point calculations.
11. **The first local correction (proof sketched).** *Source:* the manuscript, Section 28, display (28.9) and its run-family form (28.10). *Claim:* uniformly for $y = (k - m_r)/\sqrt{rv}$ in a fixed compact interval, $\mathbb{P}(X_r = k)$ equals $\varphi(y)/\sqrt{rv}$ times $1 + r^{-1}\{\beta H_2(y)/(2v) + \kappa_4(Z)H_4(y)/(24v^2)\} + O_w(r^{-3/2})$. *Sketch given:* expand $\log \mathbb{E} e^{itX_r}$ through (28.6) as $r\chi(it) + \log g_\epsilon(it)$, with $\chi''(0) = v$, $\chi^{(4)}(0) = \kappa_4(Z)$, vanishing odd derivatives of χ , $(\log g_\epsilon)''(0) = \beta$, and invert as in the proof of Theorem 28.1. *Missing:* the expansion of the characteristic function to fourth order with a remainder that is uniform and integrable after Fourier inversion; the control, at relative order $r^{-3/2}$, of the neighbourhood of zero and of its complement (where only the finite geometric sum is available); and the odd-depth case through g_1 . *Evidence at the write* (6 October 2026, from the exact histograms of the positive formula, even depths, run family $a = 3, 4, 5$, $r = 16, \dots, 256$, all k with $|y| \leq 2$): the largest relative deviation of $\sqrt{rv} \mathbb{P}(X_r = k)/\varphi(y)$ from 1 times r stays between 0.49 and 1.28, while after the correction the largest deviation times $r^{3/2}$ stays below 0.24 and decreases in r for each a . This is consistent with the display and its constants but is not a proof.
12. **Every fixed correction order (stated without proof).** *Source:* the manuscript, Section 28, after (28.9): “the same analytic expansion gives every fixed local correction order” (for fixed orders only, with no claim that the infinite expansion converges). *Missing:* a statement of the polynomials multiplying $r^{-i/2}$, $i \leq j$, in terms of the derivatives of χ and $\log g_\epsilon$ and the uniform remainder estimate $O_{w,j}(r^{-(j+1)/2})$ after the terms through $r^{-j/2}$; the case $j = 2$ is Question 11.

36 Status of the results and source record

The selected predecessor explicitly asks for sharp normalization in its critical and supercritical regimes. This article provides complete conventional proofs of the critical logarithmic constants and the supercritical amplitude and cluster interval for that same two-parameter family. The critical result is strengthened to arbitrary arrangements with the stated letter counts.

The positive histogram identity is the central exact statement. The random-sum representation, finite moments, local expansions, pressure, and extremal arrangement comparisons follow

from it with the probability arguments supplied above. The profile realization theorem supplies a separate combinatorial classification and the A001787 connection. The supercritical result uses a different proof: an explicit contraction and a branchwise invariant concave envelope.

The general spectral explanation for discrepancy growth, the existence of a unit-modulus-eigenvalue central limit theorem, and the general invariant-section identity are established literature. We have cited them and do not present them as new general theorems. The exact evaluated results in this article were derived for the present project and were not found in the targeted sources surveyed. That statement does not establish global priority.

The source snapshot used for direct file inspection was

112d6bedeb346034e89d1b1c423371d76e68fb09.

The report and its adjacent README were read at the path of this report in that snapshot (the manuscript's reference to "ProveIt: Sharp discrepancy for two families of binary substitutions, Part II, question 7 and accompanying README", inspected 5 October 2026). Search indexing also exposed the earlier commit 52d8ca404b076c38eb7c513569592ebd8c479115; the directly inspected snapshot, rather than search-index freshness, defines the predecessor addressed here. The research search was targeted to this report, OEIS entries, and the cited primary substitution literature. It was not an exhaustive audit of every issue, branch, paper, or unpublished argument related to the repository.

All prefix conventions, logarithm bases, and changes of eigenweight have been stated explicitly. Finite computations were performed with independent integer and rational checks, and asymptotic claims have analytic proofs in the text. The manuscript has not been externally refereed or proof-assistant verified. [Added 6 October 2026, batch 114: with two exceptions, recorded as Questions 11 and 12: the first local correction (28.9) and the remark on every fixed correction order are argued in outline only.]

Appendix A belongs to Part I (and is printed as it was); Appendices B and C belong to Part II.

A Reproducing and inspecting the deliverables

All verification code uses only the Python standard library and requires Python 3.10 or later. From the archive’s top-level directory run:

```
python code/independent_check.py
python code/verify.py --long
```

The shorter command `python code/verify.py` omits the 20-million-letter literal construction but retains the exact first-counterexample search and finite certificate. A successful run writes its own report and regenerates the exact data. Running the shorter command will replace the saved report with one that accurately records that the long test was omitted.

For an individual large-index query:

```
from substitution import Substitution

model = Substitution(3)
assert model.select(1, 2977771) == 5334043
assert model.select(0, 8933313) == 20222898
print(model.first_at_least(1, threshold=2))
```

Run this from `code/`, or add that directory to the Python module search path. Ranks and returned positions are 1-based; prefix lengths and internal offsets are 0-based. The exact search’s `None` result means that no witness was found through the specified maximum block level, not that the infinite sequence has no witness.

The main archive components are described in Table 3. The pre-rendered figure is included, so compiling the article does not require Python or Matplotlib. To rebuild the PDF, run `pdflatex` on `article.tex` twice; the bibliography is embedded in the source. To rebuild only the plot, run `python code/make_figure.py`, which requires Matplotlib. No fonts, checksum files, temporary TeX build files, or downloaded copies of other authors’ papers are included.

File or directory	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete source and compiled article.
<code>code/substitution.py</code>	Exact quadratic arithmetic, summaries, counts, selection, first-threshold search.
<code>code/independent_check.py</code>	Separate compact checker for the finite certificate.
<code>code/verify.py</code>	Executable cross-checks and data generation.
<code>data/certificates.json</code>	First-counterexample traces and exact certificate coefficients.
<code>data/finite_extrema.json</code>	Block extrema through W_{30} .
<code>data/extremal_family.json</code>	First 25 explicit extremal-family terms.
<code>data/verification.json</code>	Report of the executed test suite.
<code>figures/</code>	The figure in vector PDF and raster PNG form.
<code>oeis_proposed_updates.txt</code>	Suggested mathematical corrections, not submitted to OEIS.
<code>README.md</code>	Build instructions, scope, and verification notes.

Table 3: Archive contents.

B A compact proof checklist

For ease of independent checking, the argument can be reduced to the following algebraic chain.

1. $q^2 + (b + 1)q = a$ and $X = \#0 - q\#1$ imply $X(\tau(P)) = -qX(P)$.
2. Legal internal prefixes give exactly (14.4)–(14.5).
3. Since $a - bq = q(q + 1) > 0$, the extreme state-1 digits are 0 and $a - q$.
4. The extremum operator is q -contractive, and

$$(-q(\widehat{C} + 1), \widehat{C} - 1, -qC, \widehat{C})$$

is its fixed point because $\widehat{C}(1 - q^2) = a - q$ and $\widehat{C} - q^2(\widehat{C} + 1) = qb > 0$.

5. Immediately before the n th one and zero, the error maps are $q - X$ and $(X + 1)/q$, respectively.
6. The two-step extremal cycle has affine map $x \mapsto q^2x + a - q$, proving sharpness and the phase transition.

C Reproducibility files

The archive delivered with the manuscript contained the files below. The manuscript's `article.tex`, `article.pdf`, `README.md` and the file list `MANIFEST.txt` are not shipped (the text is Part II of this report; its checksum ledger was verified and retired). The other files are shipped byte-identical under the prefix `02-morphic-`:

shipped file	purpose
<code>code/02-morphic-verify.py</code>	exact standard-library verifier
<code>code/02-morphic-make_figures.py</code>	deterministic figure generator
<code>code/02-morphic-build.sh</code>	the manuscript's PDF build script
<code>data/02-morphic-verification.json</code>	recorded exact-check output
<code>figures/02-morphic-A28436*_errors.{pdf,png}</code>	vector and raster figures
<code>02-morphic-oeis_proposed_updates.txt</code>	concise draft comments, not submitted
<code>02-morphic-research_status.md</code>	novelty and verification boundaries

The programs still use their delivery names: the verifier writes `data/verification.json`, which in this report is *Part I*'s recorded report; the figure script writes unprefix `figures/A28436*_errors.*`; and the build script builds `article.tex`, which here is the whole report. Run them on a copy of the delivered layout (see the README).

References

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- [3] The OEIS Foundation Inc., *A284366: Positions of 1 in A284364; complement of A284365*. Entry credited to Clark Kimberling, March 26, 2017; density conjecture comment by Amiram Eldar, April 13, 2025. <https://oeis.org/A284366>. Accessed September 19, 2026.
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- [8] The OEIS Foundation Inc., *A184485: Upper s -Wythoff sequence, $s(n) = 3n - 2$* , <https://oeis.org/A184485>, accessed October 1, 2026.
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